

MARKETMIND AI

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MARKETMIND AI

AI-Powered Business Intelligence
and Decision Support Platform

TURNING DATA INTO OPPORTUNITIES

FINAL PROJECT DOCUMENTATION

Technical Report on System Architecture, Artificial Intelligence,
Machine Learning, Application Development and Testing

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1. Project Information

Project Details

Project Attribute	Details
Project Title	MarketMind AI: AI-Powered Business Intelligence and Decision Support Platform
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Project Type	Final Milestone Project
Program	Springboard Program
Project Domain	Artificial Intelligence, Machine Learning and Business Intelligence

2. Abstract

MarketMind AI is an AI-powered business intelligence and decision support platform designed to transform business and customer data into meaningful and actionable insights. The application combines interactive dashboards, backend services, database operations and machine learning workflows within a unified system.

The platform includes multiple analytical and intelligent capabilities, including customer segmentation, forecasting, revenue analysis, customer churn prediction, anomaly detection and product recommendations. These capabilities are designed to help users identify patterns, understand customer behaviour and support data-driven decision-making.

In addition to the AI and analytics modules, MarketMind AI includes interactive dashboards, authentication, guided user workflows, chatbot functionality and frontend visualisations. The application follows a full-stack architecture in which the frontend communicates with backend APIs, while backend services manage application logic, database operations and AI and machine learning workflows.

The generated predictions, insights and recommendations are returned to the application interface and presented through dashboards and other interactive components. This document presents the technical design, implementation, testing and current status of MarketMind AI, including its technology stack, system architecture, AI and machine learning implementation, database integration, user interface, testing, challenges, limitations and future scope.

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3. Introduction

3.1 Background

Businesses generate large amounts of data through customer transactions, sales, revenue activities, inventory, and daily operations. Although this data contains valuable information, analysing it manually can be time-consuming and may make it difficult to identify important patterns.

Traditional dashboards mainly present historical information, while businesses increasingly require systems that can forecast trends, identify unusual activity, understand customer behaviour, and provide actionable recommendations.

Artificial intelligence and machine learning can enhance business analytics by processing historical and operational data to generate predictions and insights. MarketMind AI was developed as an integrated business intelligence and decision-support platform that combines analytics, machine learning, backend services, and interactive dashboards within a unified application.

The system supports customer analysis, business performance monitoring, predictive analytics, and operational insights through accessible application features.

3.2 Problem Statement

Businesses often manage customer, sales, product, inventory, and revenue information across multiple systems or datasets. Converting this raw information into meaningful insights can be challenging, particularly when analysis depends on manual processes or separate tools.

The major challenges addressed by MarketMind AI include:

- Difficulty in analysing large volumes of business and customer data.
- Limited visibility into customer behaviour and segments.
- Difficulty in forecasting future business trends.
- Challenges in analysing revenue and performance patterns.

- Limited ability to identify potential churn risks.
- Difficulty in detecting unusual business activity.
- Limited availability of personalised product recommendations.
- Fragmented analytical workflows across multiple tools.

MarketMind AI addresses these challenges by integrating business analytics, machine learning, and interactive dashboards within a single application.

3.3 Objectives

The primary objective of MarketMind AI is to develop an integrated AI-powered business intelligence platform that analyses business and customer data and provides useful insights for decision-making.

The major objectives are:

1. To analyse customer and business data to identify meaningful behavioural and performance patterns.
2. To implement customer segmentation based on relevant characteristics and behaviour.
3. To develop forecasting and revenue analysis capabilities using historical business data.
4. To implement churn prediction for identifying customers who may be at risk of disengagement.
5. To detect anomalies and unusual patterns in business data.
6. To develop a product recommendation system using available customer and product information.
7. To provide interactive dashboards and visualisations for presenting analytical insights.
8. To integrate frontend, backend, database, authentication, and AI/ML components into a unified full-stack application.
9. To test and validate the major application and analytical components.

4. Scope

4.1 In Scope

The scope of MarketMind AI covers the development and integration of business intelligence, application, and AI/ML functionality.

The project includes:

- Interactive business dashboards and visualisations.
- Customer and business data analysis.
- Customer segmentation and behavioural analysis.
- Sales forecasting and revenue analysis.
- Customer churn prediction.
- Anomaly detection.
- Product recommendation functionality.
- Authentication and user access.
- Guided workflows and chatbot functionality.
- Frontend and backend API development.
- Database integration and data management.
- AI/ML workflow integration.
- Application testing and validation.
- Git and GitHub-based collaborative development.
- Deployment of the frontend and backend application services.

4.2 Out of Scope

The following capabilities were outside the primary scope of the current project implementation:

- Large-scale enterprise or distributed system deployment.
- A dedicated native mobile application.
- Real-time streaming analytics for continuously incoming data.
- Fully automated model retraining and production ML monitoring.
- Advanced enterprise-level access management beyond the implemented authentication and role-based controls.
- Large-scale integration with external enterprise business systems or commercial data platforms.

The project primarily focuses on demonstrating the integration of business analytics, machine learning, frontend, backend, database, and deployment functionality within the MarketMind AI platform.

5. Existing System & Proposed Solution

5.1 Existing System

Businesses commonly use spreadsheets, reporting tools, dashboards, and separate analytical applications to monitor performance. While these approaches provide useful historical information, they may require multiple tools for forecasting, customer analysis, anomaly detection, and recommendations.

The major limitations of traditional approaches include:

- Business data and analytical workflows may be distributed across multiple systems.
- Manual analysis can be time-consuming and difficult to scale.
- Traditional dashboards may provide limited predictive capabilities.
- Customer behaviour and churn risks may be difficult to identify without advanced analytics.
- Anomaly detection and personalised recommendations may require separate analytical tools.
- Direct use of analytical and machine learning tools may require technical knowledge.

These limitations create a need for an integrated platform that combines business analytics, machine learning, and accessible user interfaces.

5.2 Proposed Solution

MarketMind AI is an integrated AI-powered business intelligence and decision-support platform. It combines frontend interfaces, backend APIs, database services, analytical workflows, and machine learning components within a unified application.

The platform provides interactive dashboards and analytical capabilities for customer segmentation, forecasting, revenue analysis, churn prediction, anomaly detection, and product recommendations.

Business data is processed through backend and analytical services, and the resulting insights and predictions are presented to users through dashboards and application interfaces. This reduces the need for separate tools and provides multiple business intelligence capabilities within a single system.

5.3 Key Features

The major features of MarketMind AI are summarised below.

Feature	Description
Interactive Dashboards	Provides visual access to business information, KPIs, and analytical insights.
Customer Segmentation	Groups customers based on relevant characteristics and behavioural patterns.
Forecasting	Uses historical information to estimate future business trends.
Revenue Analysis	Provides insights into revenue patterns and business performance.
Churn Prediction	Identifies customers who may be at higher risk of disengagement.
Anomaly Detection	Identifies unusual patterns or observations requiring attention.
Product Recommendation	Generates relevant product suggestions using customer and product signals.
Authentication	Provides controlled access to application functionality.
Guided Workflows	Helps users navigate and interact with major application features.
Chatbot	Provides interactive user assistance and application interaction.
Frontend–Backend Integration	Connects the user interface with APIs, database services, and analytical workflows.

6. Technology Stack

MarketMind AI uses a modern full-stack architecture combining frontend technologies, backend APIs, database services, and AI/ML components. The technology stack supports application development, data processing, predictive analytics, and interactive business intelligence.

Technology Stack Summary

The MarketMind AI technology stack follows a layered approach. React provides the user interface, FastAPI manages backend services and APIs, PostgreSQL provides persistent data storage, and the AI/ML layer processes business data to generate predictions, insights, and recommendations.

Python is used for backend and machine learning development, while React supports the interactive web interface. Pandas and NumPy support data processing, and Scikit-learn provides machine learning capabilities.

The technologies together support the major features of MarketMind AI, including dashboards, authentication, customer analytics, segmentation, forecasting, revenue prediction, churn prediction, anomaly detection, product recommendations, and chatbot functionality.

For deployment, the React frontend is hosted on Vercel and the FastAPI backend is deployed separately on Render. The backend connects to the PostgreSQL database hosted through Neon and integrates the required AI/ML workflows.

Table 6.1: Technology Stack Used in MarketMind AI

Category	Technology	Purpose
Programming Languages	Python, JavaScript	Used for backend development, AI/ML processing, data analysis, and frontend development.
Frontend	React	Used to develop the interactive user interface, dashboards, visualisations, and application features.
Backend	FastAPI, Python	Used for backend services, REST APIs, business logic, and communication between application components.
Database	PostgreSQL / Neon	Used for persistent storage and management of application data.
AI / Machine Learning	Scikit-learn and Python ML libraries	Used for segmentation, forecasting, churn prediction, anomaly detection, revenue prediction, and product recommendations.
Data Processing	Pandas, NumPy	Used for data cleaning, transformation, analysis, and feature preparation.
Data Visualisation	React-based charting and dashboard libraries	Used to present business insights, analytical results, predictions, and performance trends.
Authentication	JWT-based authentication and security mechanisms	Used to provide secure user authentication and controlled access.
API	REST APIs / FastAPI	Used for communication between the frontend, backend, database, and AI/ML services.
Version Control	Git and GitHub	Used for source code management, collaboration, branch management, and version control.
Frontend Deployment	Vercel	Used to deploy and host the React frontend application.
Backend Deployment	Render	Used to deploy and host the FastAPI backend API.

7. System Architecture

MarketMind AI follows a layered architecture that integrates the user interface, backend services, business logic, AI/ML components, analytics, database, and file storage into a unified business intelligence platform.

The architecture separates presentation, API communication, business processing, intelligence, analytics, security, and data persistence responsibilities.

High-Level Architecture

Figure 7.1 presents the layered architecture of MarketMind AI, including the users, security and authentication, React frontend, FastAPI backend, business logic, AI/ML services, analytics, database, and file storage.

Architecture Layers

The major layers of the MarketMind AI architecture are:

- **Users Layer:** Supports Business Owner, Store Manager, Sales Executive, and System Administrator roles.
- **Security and Authentication Layer:** Handles registration, login, JWT authentication, role-based access, and permissions.
- **Presentation Layer:** Provides the React frontend for dashboards, sales, inventory, customers, invoices, analytics, reports, settings, and notifications.
- **API and Backend Layer:** Provides FastAPI services for authentication, business operations, reports, file uploads, and AI/ML integration.
- **Core Processing Layer:** Handles sales processing, inventory tracking, revenue calculation, invoice processing, and business logic.
- **AI/ML Layer:** Provides forecasting, customer segmentation, churn prediction, recommendations, anomaly detection, and revenue prediction.

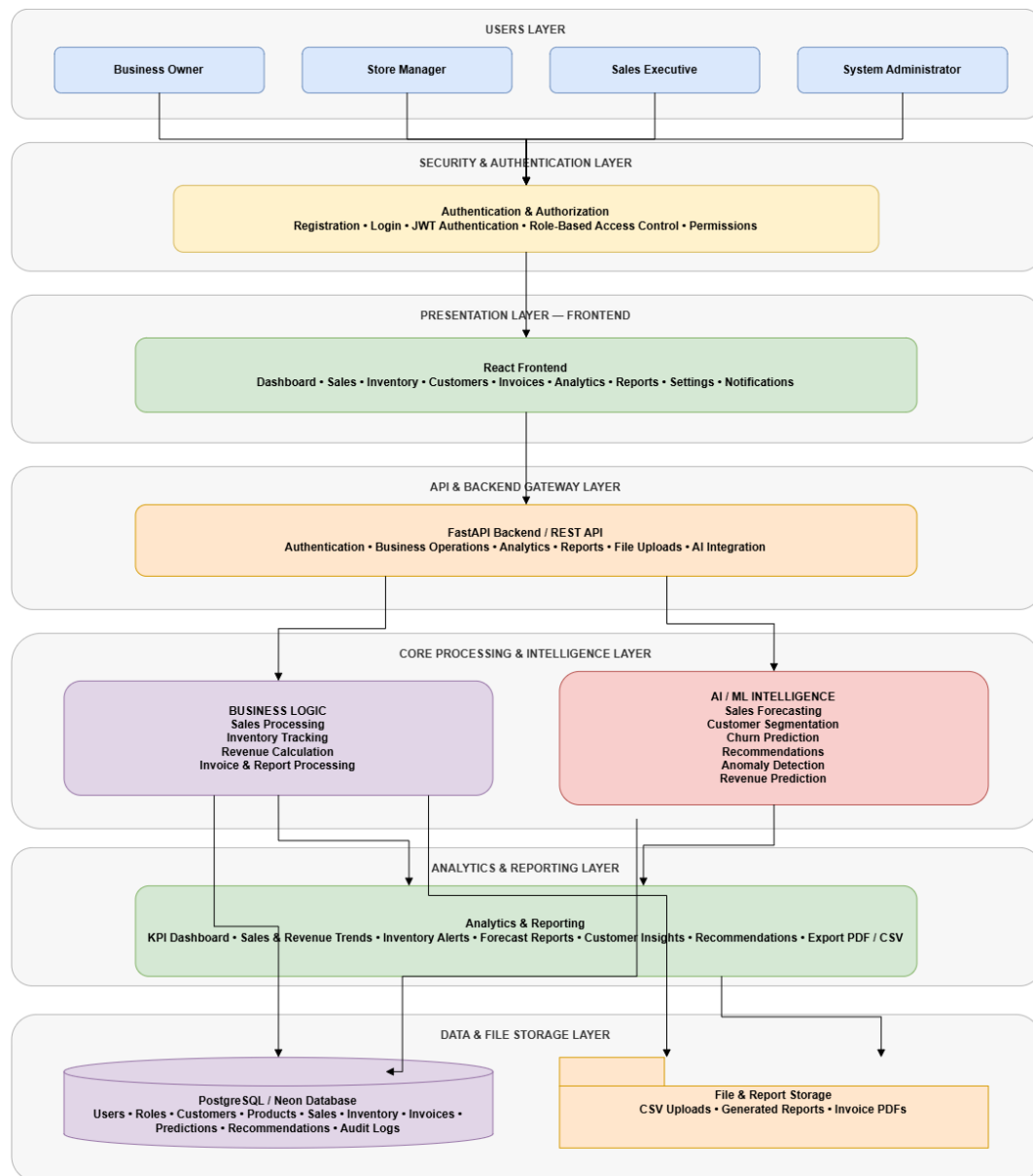


Figure 7.1: Layered System Architecture of MarketMind AI

- **Analytics and Reporting Layer:** Presents KPIs, trends, forecasts, customer insights, alerts, recommendations, and export functionality.
- **Data and File Storage Layer:** PostgreSQL stores application data, while file storage supports uploaded datasets, generated reports, and invoice PDFs.

Component Interaction

The major components communicate through a request and response workflow:

1. The user interacts with the React frontend.
2. The frontend sends requests to the FastAPI backend.
3. The backend validates and processes the request.
4. Required database or AI/ML operations are performed.
5. The processed result is returned through the backend.
6. The frontend presents the result through dashboards, visualisations, reports, alerts, or recommendations.

This architecture enables MarketMind AI to combine business operations, data management, analytics, and machine learning within a single integrated platform.

7.1 System Modules

MarketMind AI consists of interconnected modules supporting business operations, data management, customer intelligence, predictive analytics, and decision support.

Module	Description
Dashboard	Provides an overview of business information, KPIs, and analytical insights.
Sales and Invoice Management	Manages sales records and invoice-related information.
Inventory Management	Supports product and inventory management.
Customer Management	Manages customer information and supports customer analytics.

Module	Description
Categories and Suppliers	Organises product categories and supplier information.
Team Management	Supports application user and team management.
Dataset Management	Supports datasets used for analysis and AI/ML processing.
Forecasting and Revenue Analysis	Analyses historical data to generate forecasts and revenue insights.
Customer Segmentation	Groups customers according to relevant characteristics and behaviour.
Churn Risk Analysis	Identifies customers who may be at risk of churn.
Product Recommendation System	Generates relevant recommendations using customer and product data.
Anomaly Detection	Identifies unusual patterns in business data.
Chatbot and Guided Features	Provides interactive assistance and guided application workflows.
Settings and Configuration	Provides application-level settings and configuration functionality.

8. AI/ML Implementation

MarketMind AI integrates machine learning and analytical techniques to support customer intelligence, predictive analytics, and business decision-making. The major AI/ML capabilities include customer segmentation, sales forecasting, revenue analysis, churn prediction, anomaly detection, and product recommendations.

8.1 Dataset

The system uses business-related data covering customers, products, sales, transactions, invoices, inventory, and historical business performance. Depending on the analytical module, relevant customer, transactional, product, and historical data is used to identify patterns, generate predictions, detect anomalies, and produce business insights.

The available data may include:

- Customer information and behavioural data.
- Sales and transaction records.
- Product and inventory information.
- Invoice and revenue information.
- Historical business performance data.
- Customer activity and engagement information.

The dataset characteristics and input features vary according to the requirements of each AI/ML module.

8.2 Data Processing

Before applying AI/ML techniques, the available data is prepared and transformed into a suitable format. The general processing stages are:

1. **Data Collection:** Relevant business and customer data is obtained from available datasets and application sources.

2. **Data Cleaning:** Missing values, duplicates, inconsistent formats, and invalid values are reviewed and handled where applicable.
3. **Data Transformation:** Raw data is converted into formats suitable for analysis and model processing.
4. **Feature Preparation:** Relevant attributes are selected or derived based on the requirements of each module.
5. **Module-Specific Processing:** Additional processing is performed for tasks such as customer analysis, time-based aggregation, transaction analysis, and product interactions.

The processed data is then provided to the appropriate analytical model or algorithm.

8.3 Model / Algorithm

Different AI/ML approaches are used according to the business problem and required output. The major components are summarised below.

Table 8.1: Major AI/ML Components in MarketMind AI

AI/ML Module		Analytical Purpose
Customer	Segmentation	Groups customers based on relevant characteristics and behavioural patterns.
	Forecasting	Uses historical trends to estimate future business values.
Revenue Analysis		Identifies revenue-related patterns, trends, and performance insights.
Churn Risk Analysis		Identifies customers who may be at risk of discontinuing engagement.
Anomaly Detection		Identifies unusual patterns and irregular behaviour in business data.
Product	Recommendation System	Uses customer, product, and behavioural information to generate relevant product recommendations.

The system therefore combines predictive, clustering, analytical, anomaly-detection, and recommendation-based approaches to support different business intelligence requirements.

8.4 Training & Evaluation

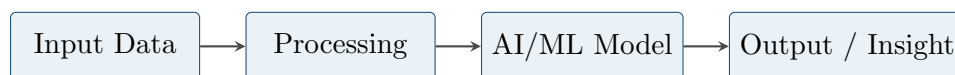
The training and evaluation process depends on the AI/ML technique being used. Predictive models are trained using relevant historical data, while clustering, anomaly detection, and recommendation components process available data according to their respective methods.

Model evaluation uses task-specific metrics where applicable. Classification models may be evaluated using accuracy, precision, recall, and F1-score, while forecasting and regression models may use MAE, RMSE, and R^2 . Clustering and anomaly-detection outputs are evaluated based on the quality and usefulness of the identified patterns.

The actual evaluation results are presented separately in Chapter 14.

8.5 AI/ML Workflow

The overall AI/ML workflow of MarketMind AI is:



Relevant business data is first processed according to the selected module. The appropriate model or algorithm then performs prediction, clustering, detection, analysis, or recommendation generation. The resulting insights are returned through the backend and presented to users through dashboards, visualisations, recommendations, alerts, and other application features.

9. Database & API

MarketMind AI uses a structured database and API layer to support data storage, retrieval, processing, and communication between the frontend, backend, and AI/ML components.

9.1 Database

The database stores the core information required for business operations, analytics, and AI/ML workflows. The main data areas include users, customers, products, inventory, sales, invoices, suppliers, categories, and datasets.

Table 9.1: Major Database Entities

Entity / Table	Purpose
Users / Team	Stores application user and team information.
Customers	Stores customer information used for management, segmentation, and churn analysis.
Products / Inventory	Stores product and inventory information for business operations and analysis.
Sales	Stores sales records used for performance analysis, revenue insights, and forecasting.
Invoices	Stores information related to business transactions.
Categories	Organises products into categories for management and analysis.
Suppliers	Stores supplier information related to products and inventory.
Datasets	Supports datasets used for analysis and AI/ML workflows.

Database Role

The database acts as the central data layer of MarketMind AI. Backend services retrieve and update data based on application requests, while AI/ML modules use relevant data

for analytical processing.

Frontend → **Backend/API** → **Database**

For analytical operations, data can be retrieved and processed through the AI/ML layer before the results are returned to the frontend.

Database Schema / ER Diagram

The ER diagram illustrates the major database tables and their relationships within the MarketMind AI application.

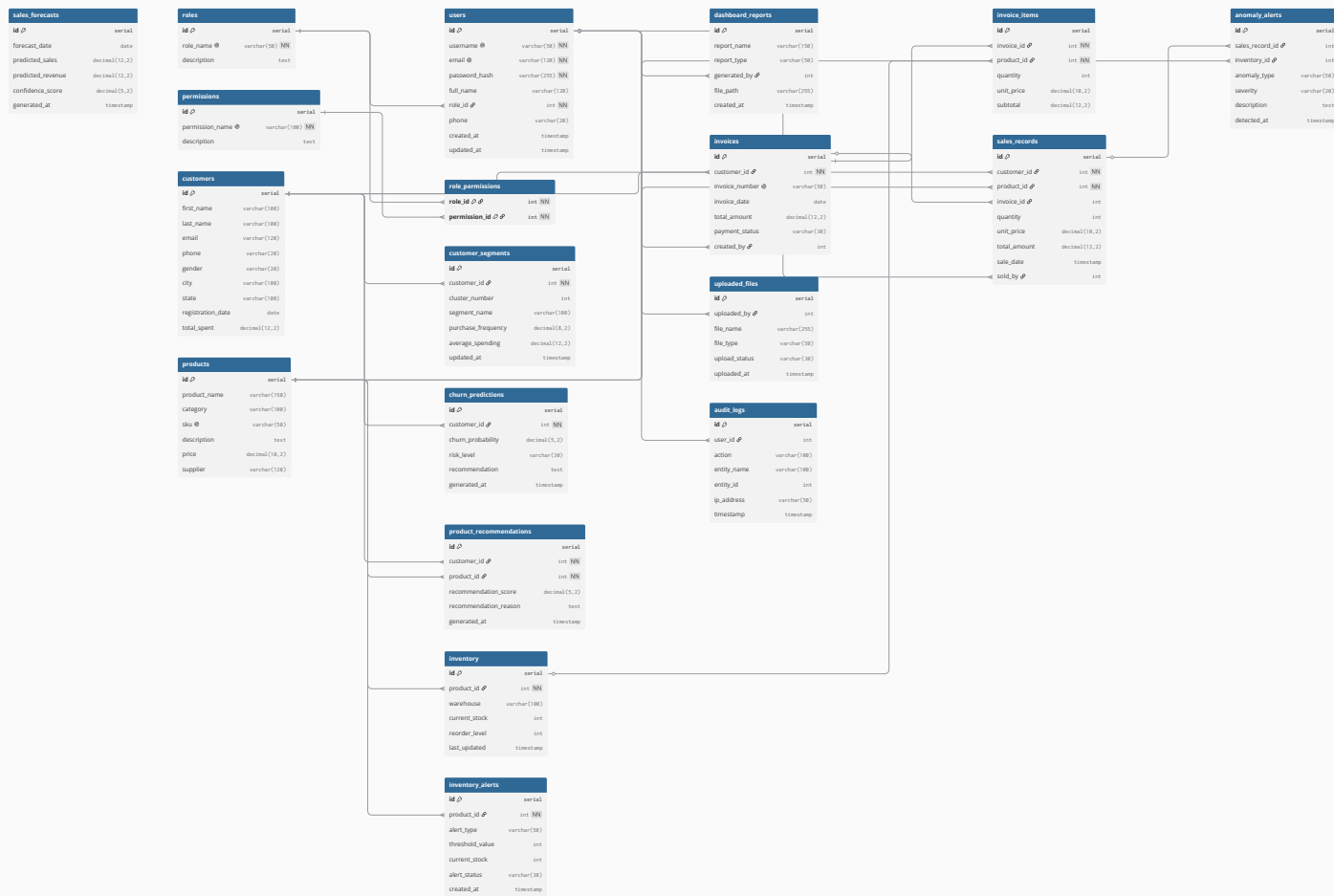


Figure 9.1: MarketMind AI Database Schema / Entity Relationship Diagram

API Communication

The API layer connects the frontend with backend services, database operations, and AI/ML functionality. When a user performs an action, the frontend sends a request to the backend, which validates and processes it before returning the result.

The general workflow is:

1. User performs an action through the frontend.
2. Frontend sends a request to the backend API.
3. Backend validates and processes the request.
4. Required database or AI/ML operations are performed.
5. Backend prepares and returns the response.
6. Frontend displays the result to the user.

API and AI/ML Integration

The API layer integrates the application's AI/ML functionality with the backend. Requests for forecasting, customer segmentation, churn prediction, anomaly detection, and product recommendations are processed through the appropriate backend workflows, with the resulting insights returned to the frontend.

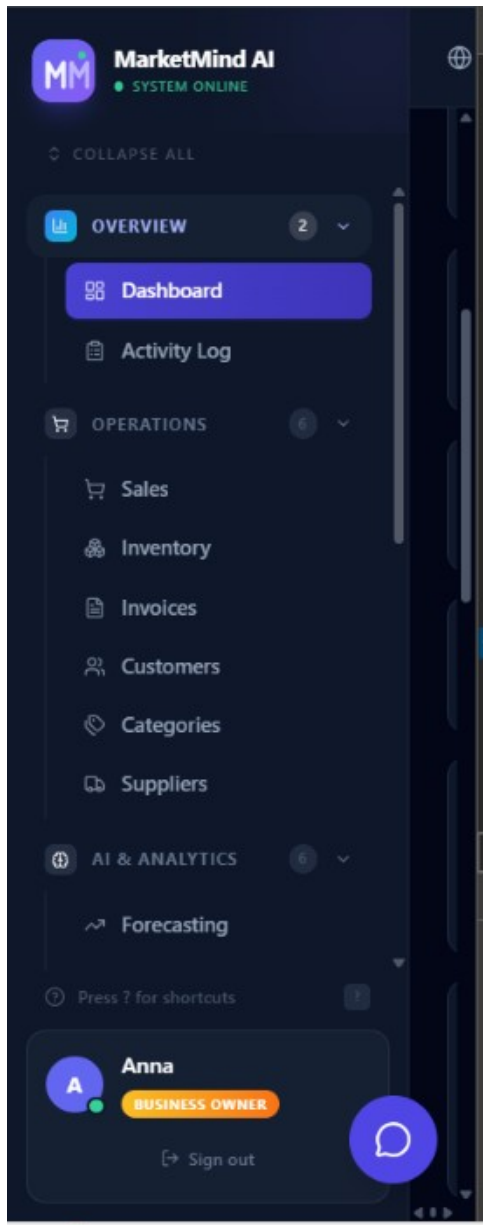
10. UI Interface

The MarketMind AI user interface provides a centralized environment for business operations, AI-driven analytics, reporting, and administration. The interface is organized into functional modules to provide easy navigation and access to business insights.

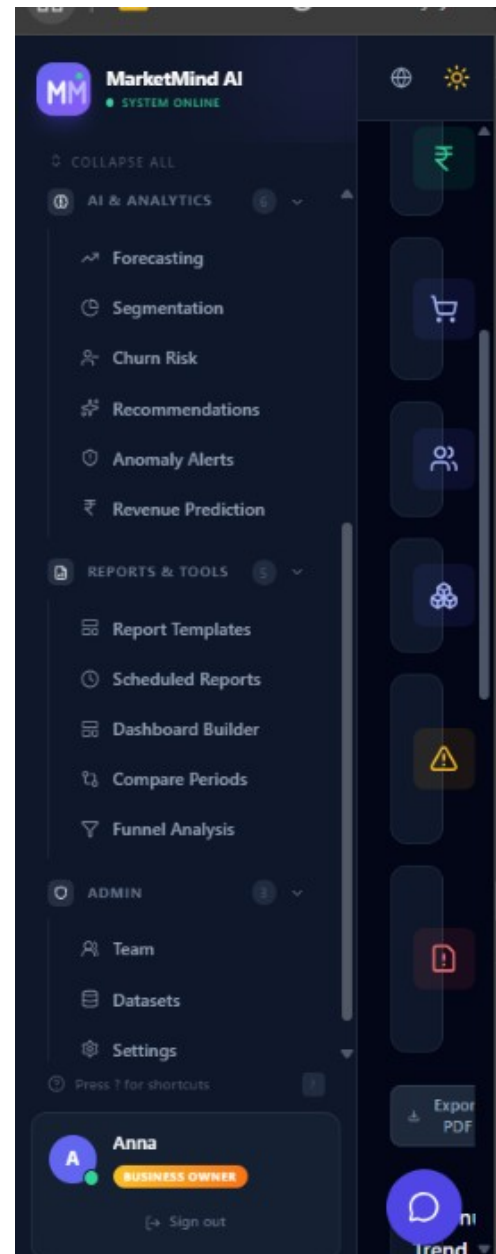
10.1 Application Navigation

The application uses a sidebar-based navigation structure that allows users to access the major business and analytical modules from a central location.

The navigation includes modules such as Dashboard, Sales, Inventory, Invoices, Customers, Categories, Suppliers, Team, Datasets, Forecasting, Segmentation, Churn Risk, Recommendations, Anomaly Alerts, and Settings.



(a) Overview and Operations navigation

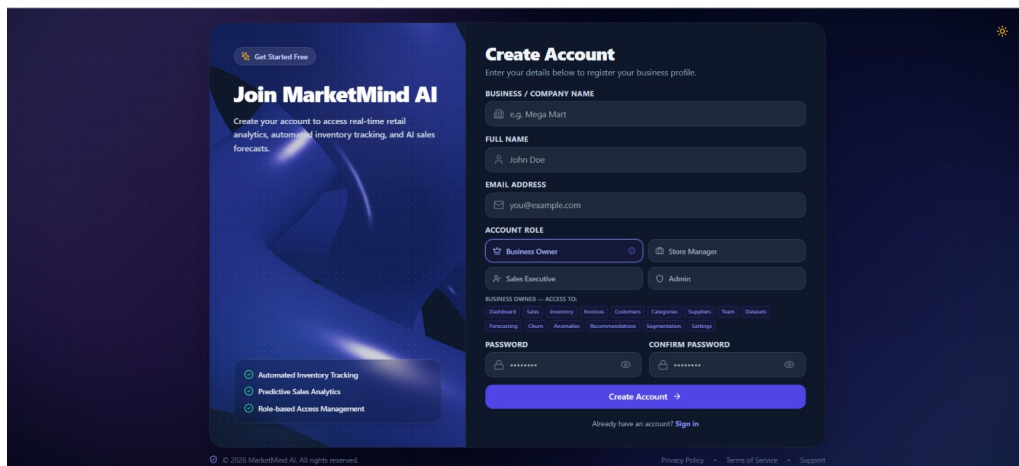


(b) AI, Reports, Tools and Admin navigation

Figure 10.1: MarketMind AI Application Navigation Sidebar

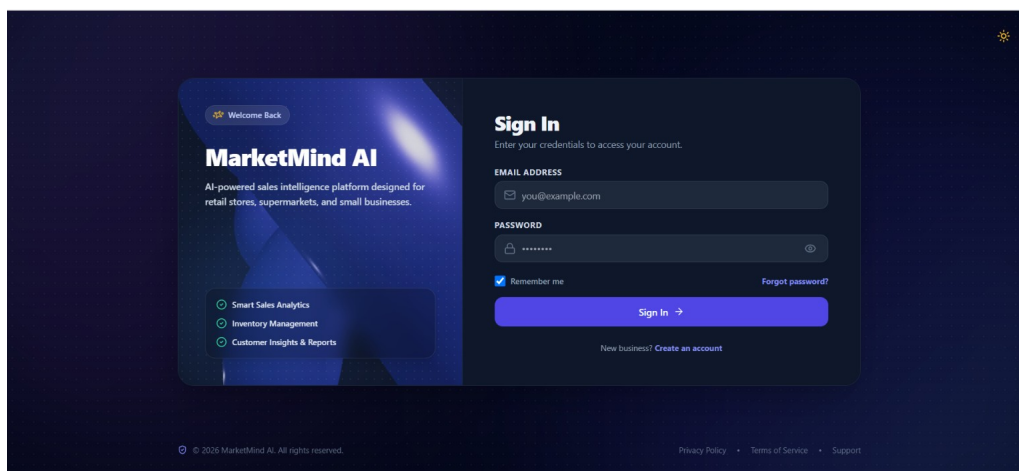
10.2 Authentication and Onboarding

MarketMind AI provides user authentication and onboarding interfaces for secure access to the platform. Users can create an account, sign in using their credentials, and follow the guided onboarding tour to understand the main features of the application.



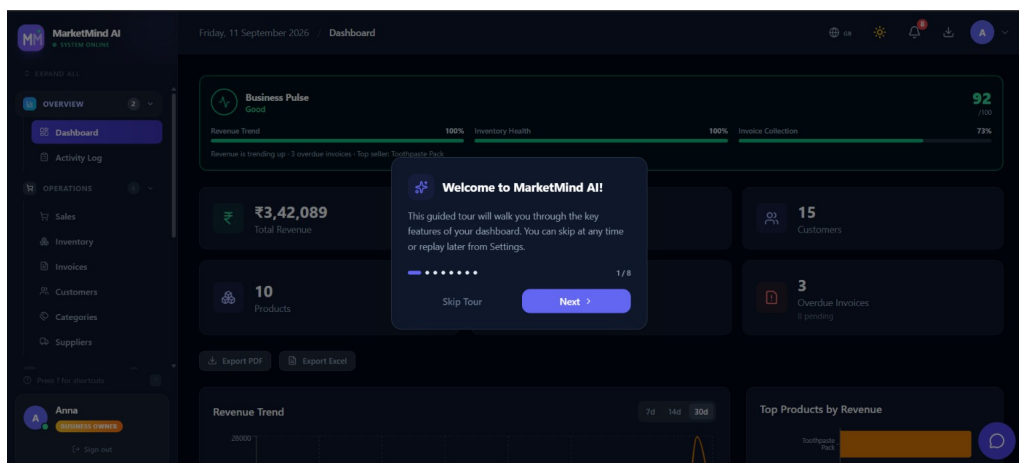
The registration form is titled "Join MarketMind AI" and "Create Account". It includes fields for "BUSINESS / COMPANY NAME" (with a placeholder "e.g. Mega Mart"), "FULL NAME" (with a placeholder "John Doe"), and "EMAIL ADDRESS" (with a placeholder "you@example.com"). Under "ACCOUNT ROLE", there are four buttons: "Business Owner", "Store Manager", "Sales Executive", and "Admin". Below these, a "BUSINESS OWNER - ACCESS TO:" section lists various features like Dashboard, Sales, Inventory, etc. There are also fields for "PASSWORD" and "CONFIRM PASSWORD", and a "Create Account" button. A link for "Already have an account? Sign In" is at the bottom.

(a) User registration and role selection



The sign-in form is titled "Sign In" and "Welcome Back". It includes fields for "EMAIL ADDRESS" (with a placeholder "you@example.com") and "PASSWORD". There is a "Remember me" checkbox and a "Forgot password?" link. A "Sign In" button is at the bottom. A link for "New business? Create an account" is also present.

(b) User sign-in interface



(c) Guided application tour

Figure 10.2: MarketMind AI Authentication and Onboarding

10.3 Dashboards

MarketMind AI provides role-based dashboards that present business information, operational metrics, and analytics according to the responsibilities of each user. The dashboard interface is customized for system administrators, business owners, store managers, and sales executives, ensuring that each role has access to the most relevant business insights and operations.

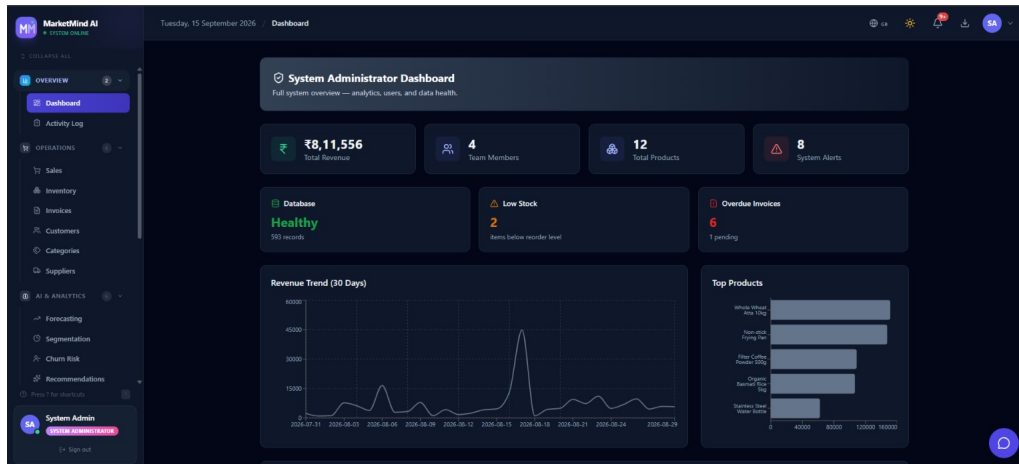


Figure 10.3: System Administrator Dashboard

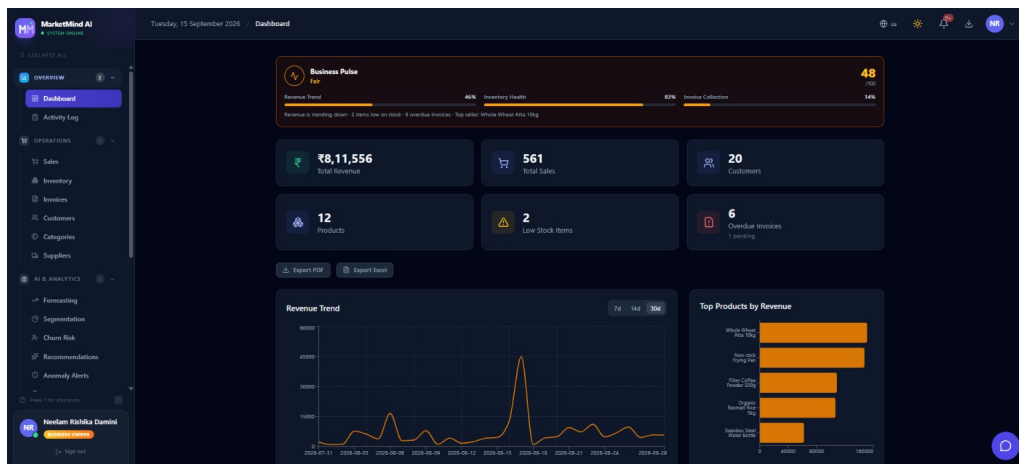


Figure 10.4: Business Owner Dashboard

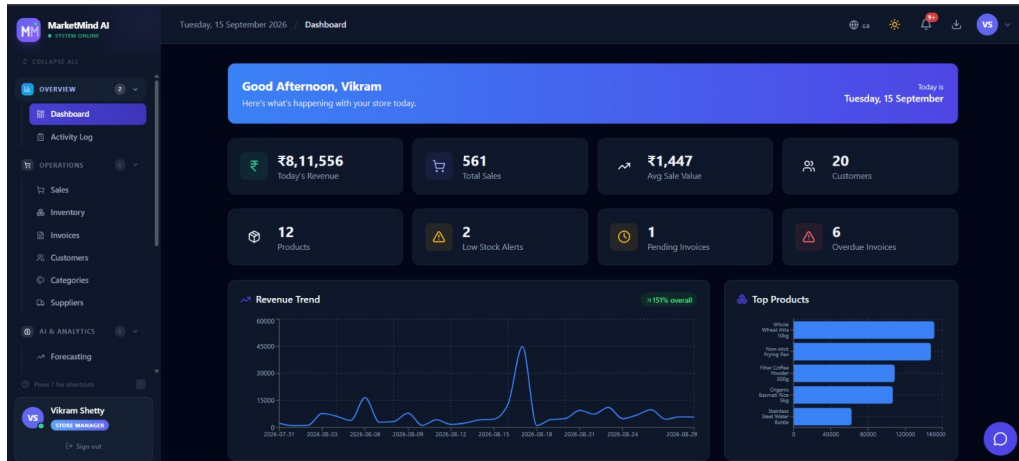


Figure 10.5: Store Manager Dashboard

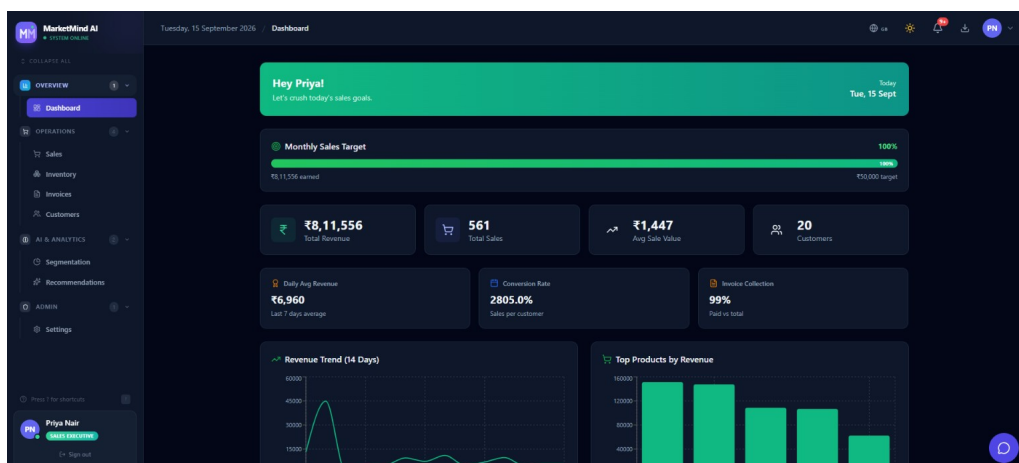


Figure 10.6: Sales Executive Dashboard

10.4 Core Business Operations

The core business modules support day-to-day management of sales, inventory, invoices, product categories, and suppliers. These interfaces allow users to monitor and manage operational data through a centralized platform.

Date	Product	Customer	Qty	Price	Total	Source
8/29/2026	Cotton T-Shirt	Vikram Singh	7	₹1,765.75	₹11,011.48	manual
8/29/2026	Toothpaste Pack	Ravi Kumar	7	₹1,255.25	₹3,363.46	manual
8/25/2026	Basmati Rice 5kg	Rohit Verma	5	₹1,177.25	₹12,544.83	manual
8/25/2026	Whole Wheat Atta 10kg	Arjun Rao	3	₹1,769.5	₹5,235.71	manual
8/25/2026	USB Cable	Priya Patel	9	₹655.97	₹10,100.87	manual
8/20/2026	Dish Soap 1L	Suresh Iyer	8	₹454.61	₹4,591.28	manual
8/19/2026	Whole Wheat Atta 10kg	Sneha Gupta	1	₹2,028.21	₹6,137.24	manual
8/18/2026	Sunflower Oil 1L	Deepak Nair	9	₹2,302.75	₹9,064.38	manual
8/17/2026	Dish Soap 1L	Deepak Nair	5	₹1,887.77	₹6,540.82	manual

Figure 10.7: Sales management interface

Name	Category	Price	Stock	Location	Actions
Pressure Cooker 5L	—	₹1,800	20	—	🔍 🔄
Dish Soap 1L	—	₹150	70	—	🔍 🔄
Cotton T-Shirt	—	₹350	50	—	🔍 🔄
Rice Cooker	—	₹2,500	15	—	🔍 🔄
USB Cable	—	₹250	40	—	🔍 🔄
Notebook 200pg	—	₹80	100	—	🔍 🔄
Toothpaste Pack	—	₹120	60	—	🔍 🔄
Sunflower Oil 1L	—	₹180	80	—	🔍 🔄
Basmati Rice 5kg	—	₹650	35	—	🔍 🔄
Whole Wheat Atta 10kg	—	₹480	50	—	🔍 🔄

Figure 10.8: Inventory management interface

Invoices
Track payment status and manage billing.

15 TOTAL, 4 PAID, 8 PENDING, 3 OVERDUE, ₹52,241.14 TOTAL AMOUNT

Invoice #	Customer	Amount	Due Date	Status	Actions
INV-SEED-005-0000	—	₹6,061.63	8/31/2026	pending	Mark Paid
INV-SEED-005-0001	—	₹6,023.53	9/4/2026	paid	
INV-SEED-005-0002	—	₹3,664.41	9/18/2026	pending	Mark Paid
INV-SEED-005-0003	—	₹2,244.96	9/6/2026	pending	Mark Paid
INV-SEED-005-0004	—	₹1,991.28	9/17/2026	pending	Mark Paid
INV-SEED-005-0005	—	₹3,869.07	9/7/2026	overdue	Mark Paid
INV-SEED-005-0006	—	₹6,543.64	9/24/2026	pending	Mark Paid

(a) Invoice management interface

Figure 10.9: Core Business Operations — Sales, Inventory and Invoices

Categories
Manage product categories.

5 TOTAL CATEGORIES, 10 TOTAL PRODUCTS, 0 ACTIVE CATEGORIES

Name	Description	Products
Clothing	—	0
Electronics	—	0
Groceries	—	0
Home & Kitchen	—	0
Personal Care	—	0

(a) Product category management

Suppliers
Manage product suppliers and vendors.

4 TOTAL SUPPLIERS

Name	Phone	Email	Address
GreenLeaf Organics	9845012345	hello@greenleaforganics.in	Bengaluru
Krishna Distributors	9876543210	sales@krishnadistributors.in	Hyderabad
Mehta Electronics	9866112233	contact@mehtaelectronics.in	Chennai
Sunrise Agro Foods	9876501234	orders@sunriseagro.in	Vijayawada

(b) Supplier management interface

Figure 10.10: Core Business Operations — Categories and Suppliers

10.5 AI and Analytics Modules

The AI and analytics section provides data-driven business insights through forecasting, customer segmentation, churn analysis, product recommendations, anomaly detection, and revenue prediction. These modules enable users to analyse business data and support data-informed decision-making.

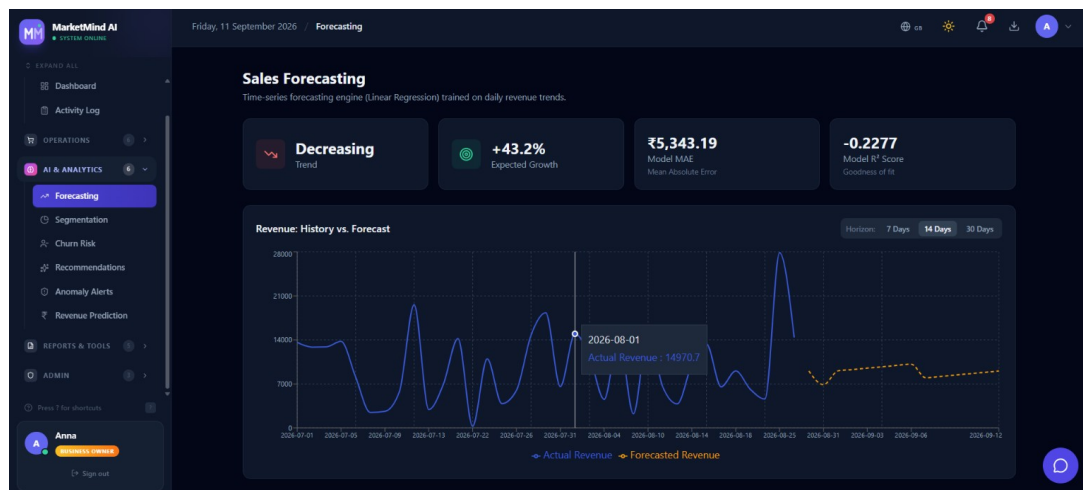


Figure 10.11: Sales forecasting interface

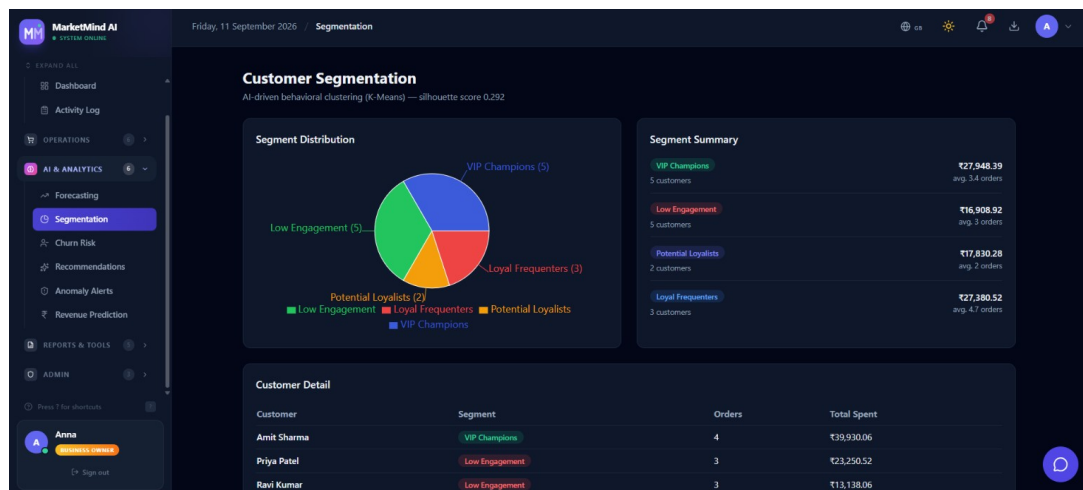


Figure 10.12: Customer segmentation interface

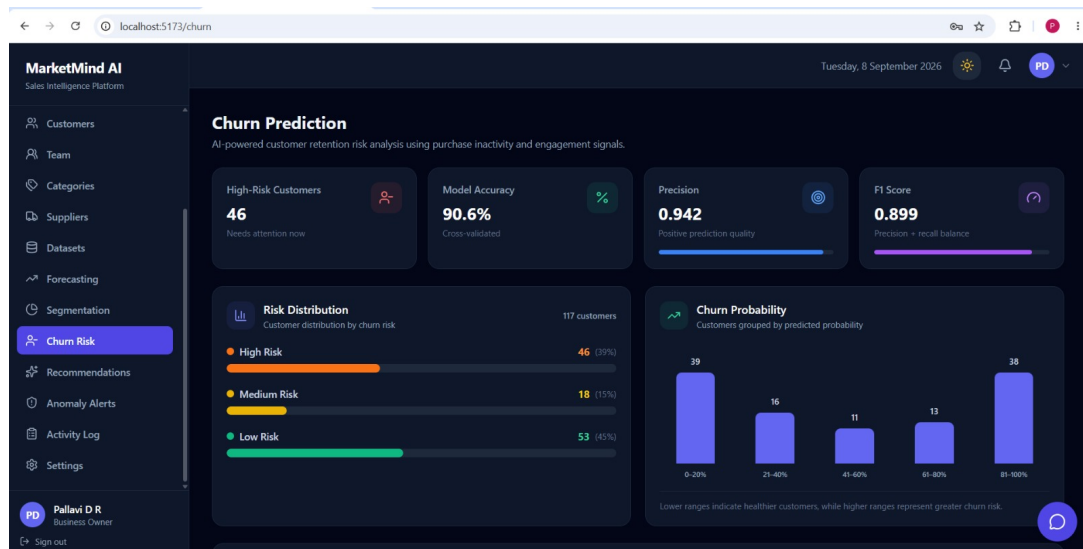
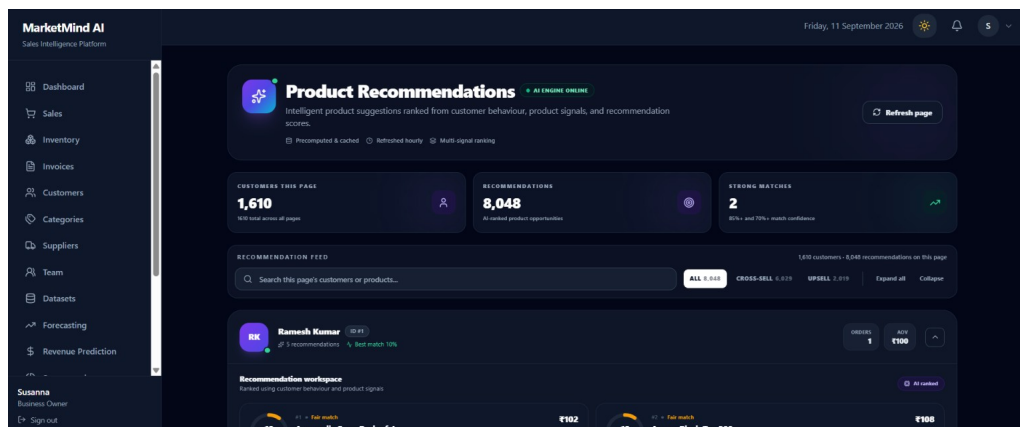
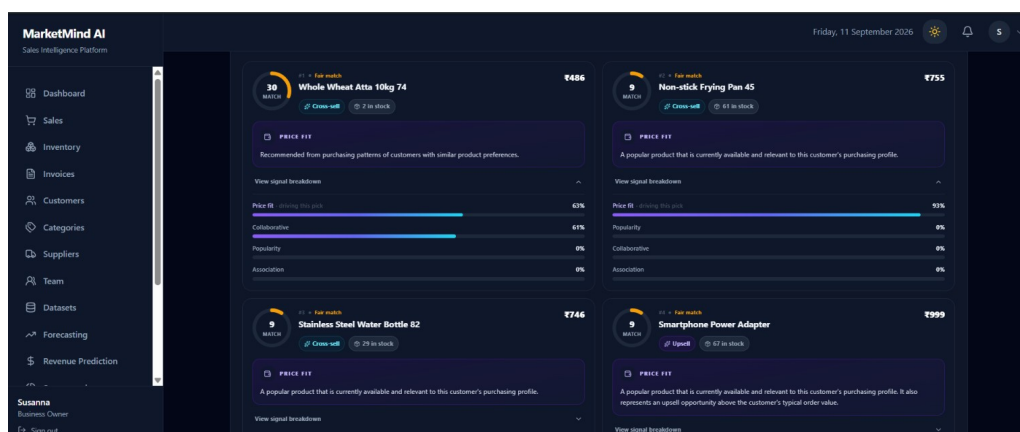


Figure 10.13: Customer churn risk analysis

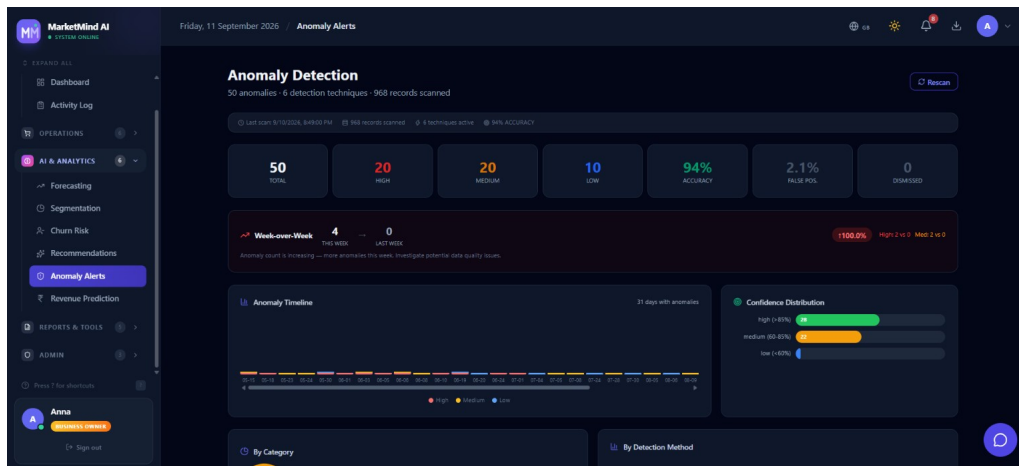


(a) Customer-level product recommendations

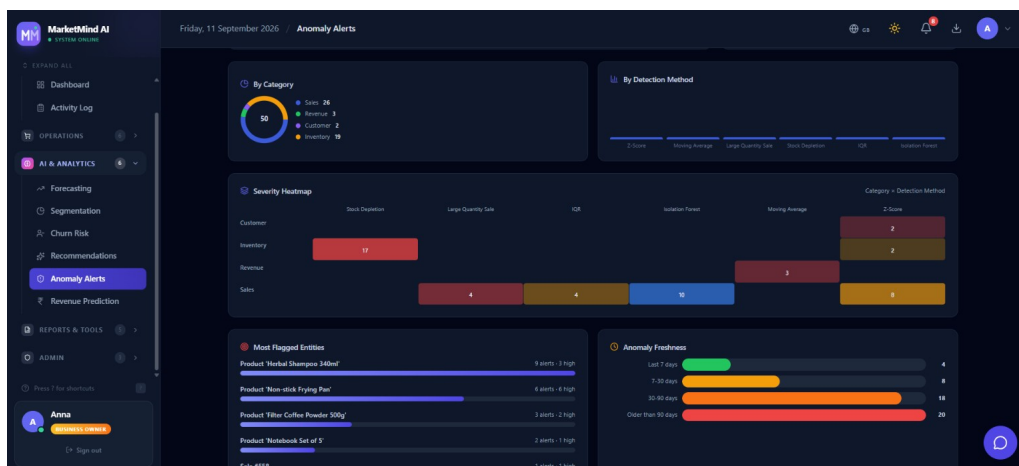


(b) Product recommendation dashboard

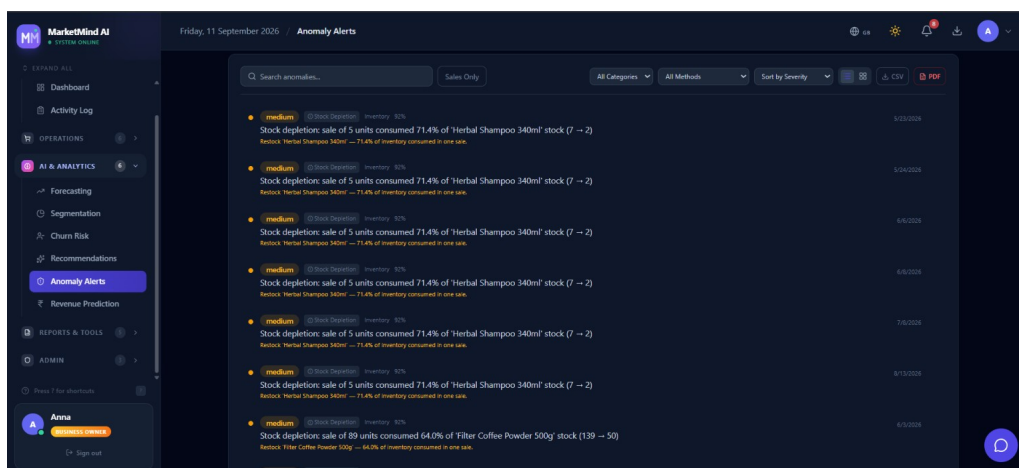
Figure 10.14: Product Recommendation Interface



(a) Anomaly alerts and severity list



(b) Anomaly analytics and detection analysis



(c) Anomaly detection dashboard

Figure 10.15: Anomaly Detection and Alerts Interface

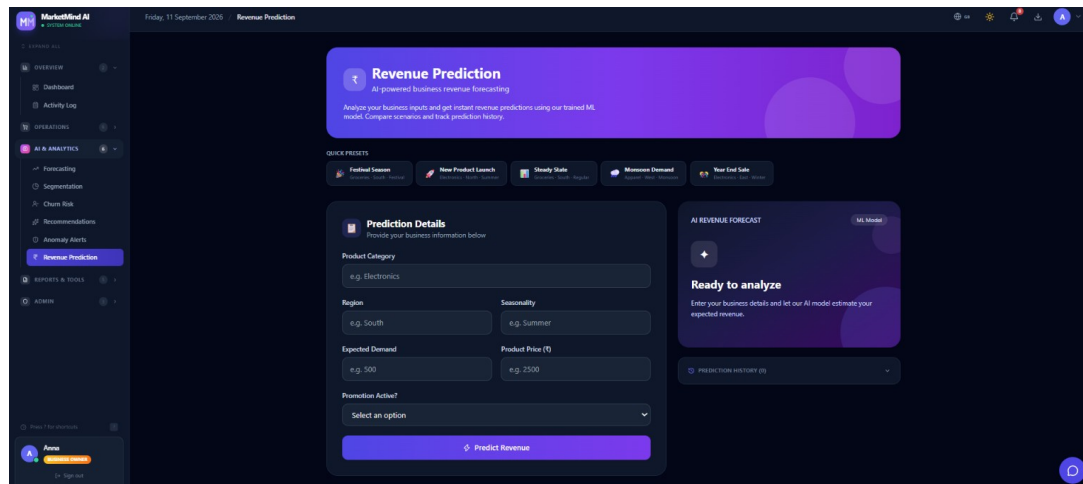


Figure 10.16: Revenue prediction interface

10.6 Reports and Business Tools

MarketMind AI provides reporting and business analysis tools that allow users to compare business performance, analyse customer conversion, build customized dashboards, and schedule recurring reports.

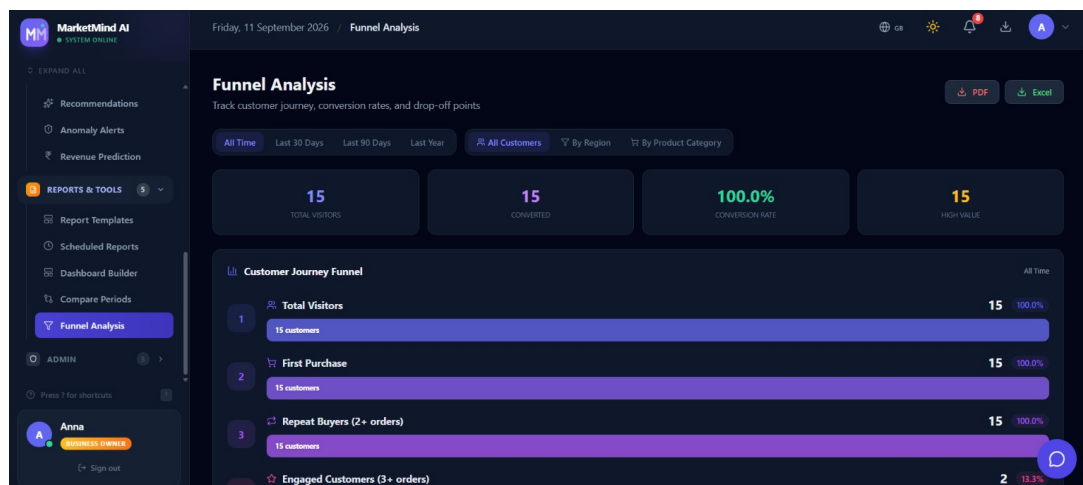
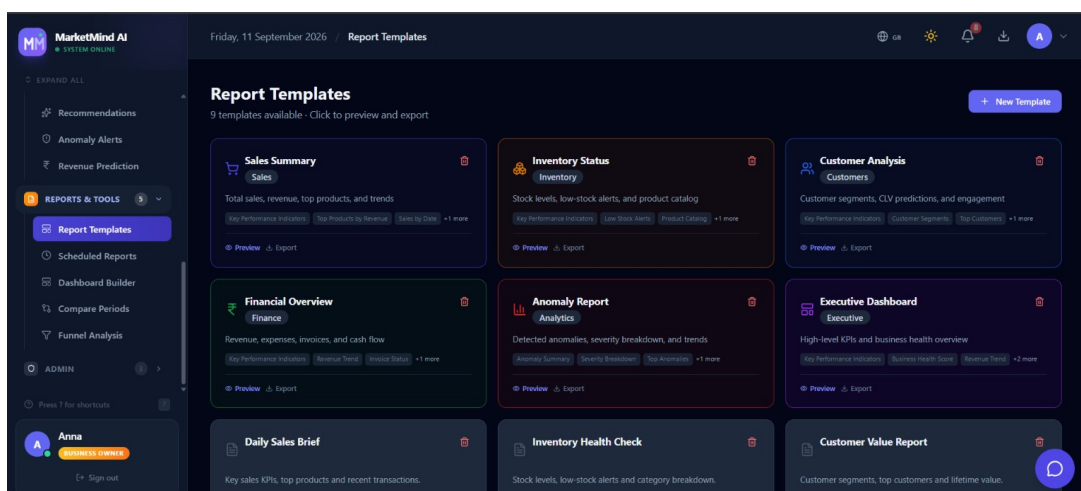
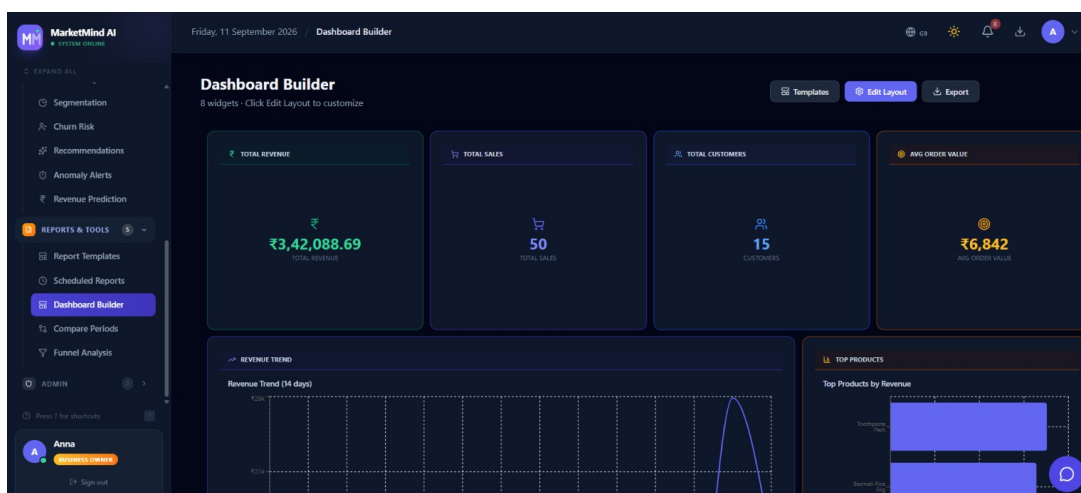
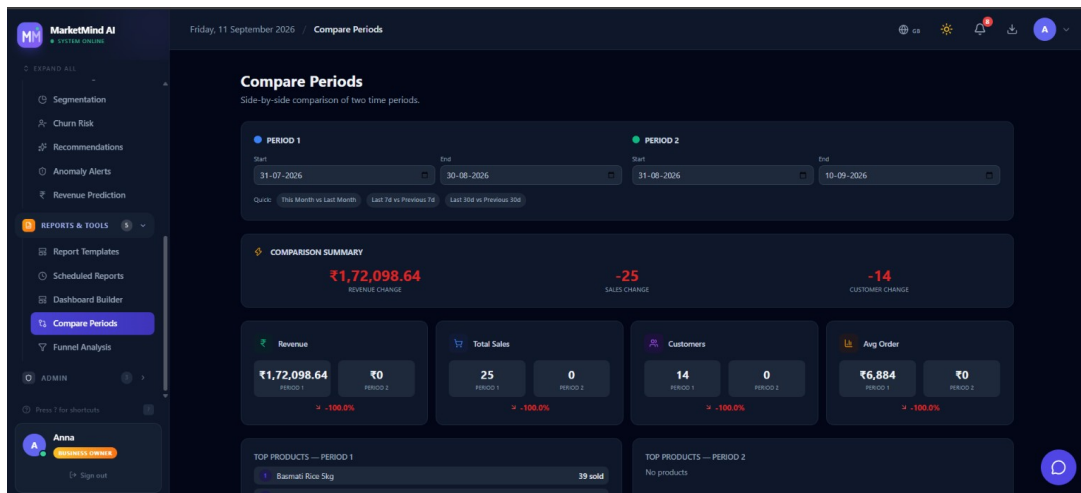


Figure 10.17: Customer journey funnel analysis



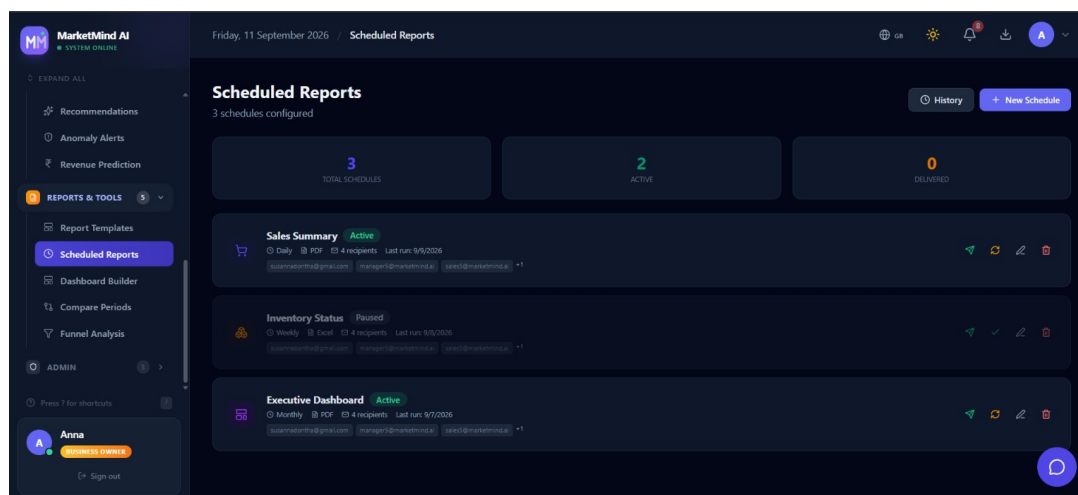


Figure 10.21: Scheduled report management

10.7 Administration and Data Management

The administration and data management interfaces provide facilities for managing datasets, user preferences, and system activity. These features support data administration, application configuration, and monitoring of user activities.

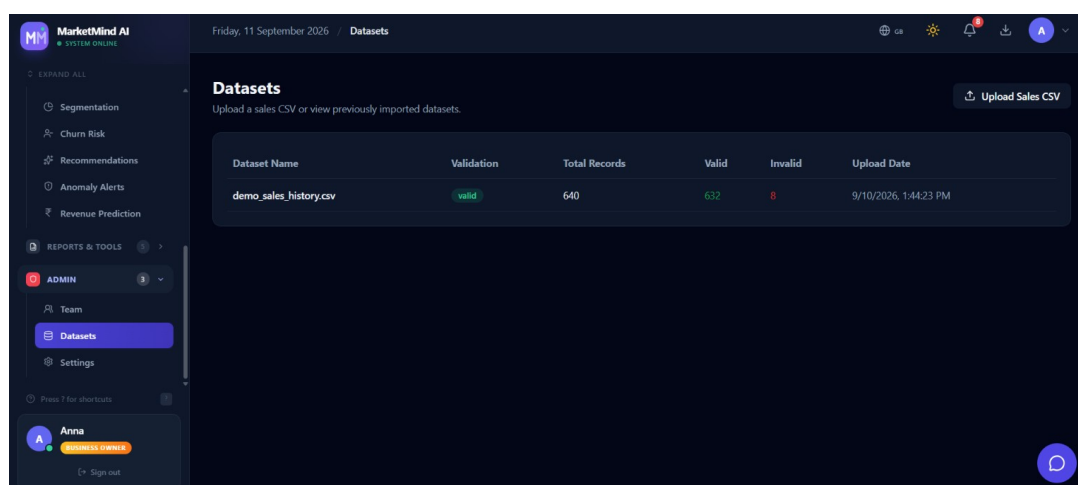


Figure 10.22: Dataset management and validation interface

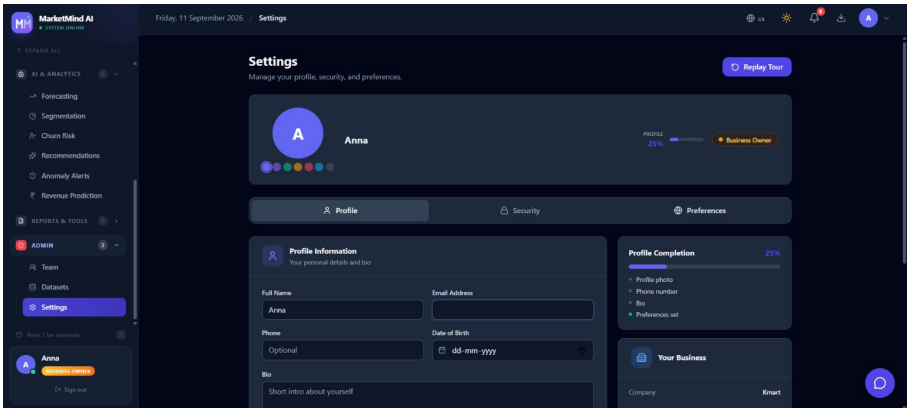


Figure 10.23: User profile, security, and preference settings

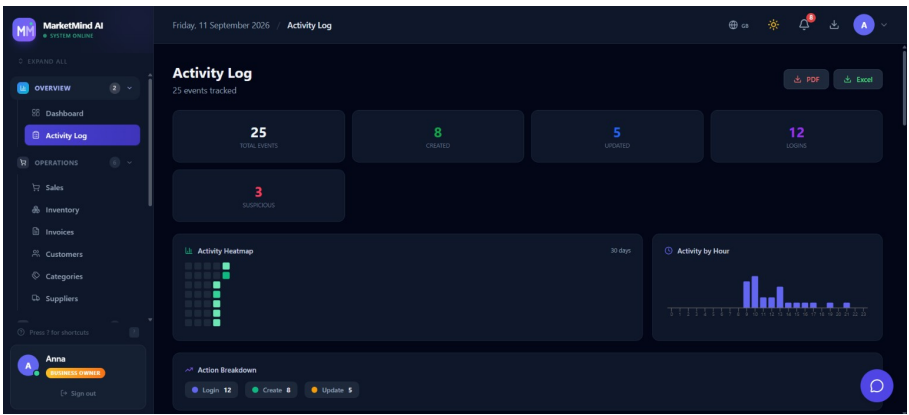


Figure 10.24: Activity summary and usage analytics

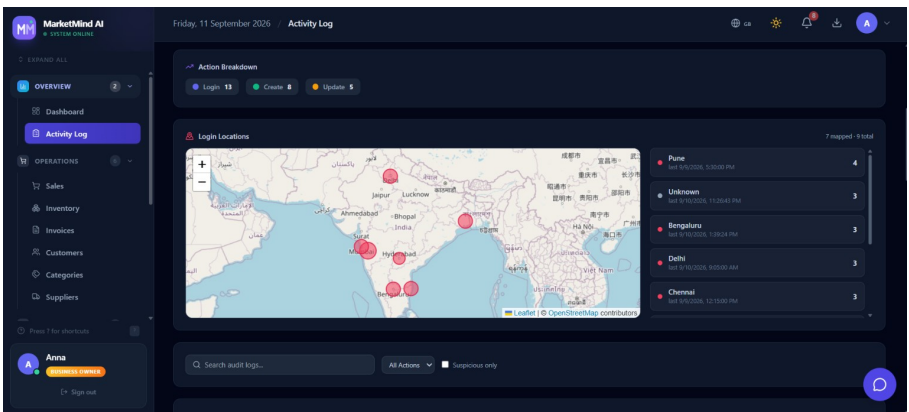


Figure 10.25: Login locations and activity monitoring

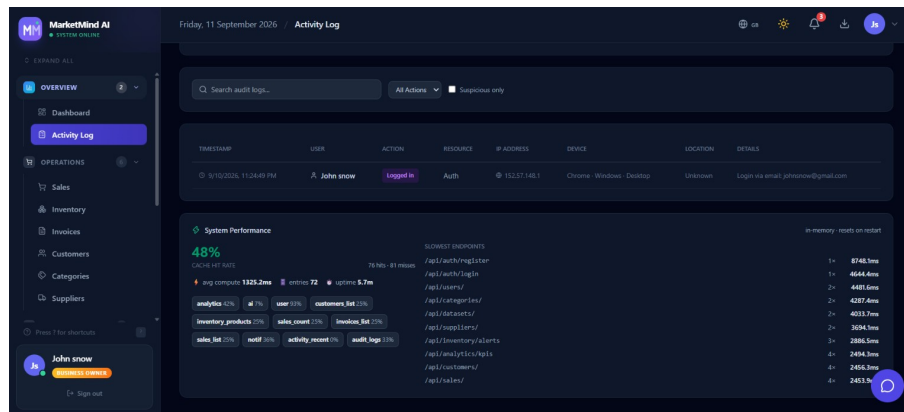


Figure 10.26: Activity events and system performance

11. GitHub & Version Control

GitHub was used as the primary platform for source code management, version control, and team collaboration during the development of MarketMind AI. A branch-based workflow allowed team members to develop different features independently while maintaining a shared codebase.

Repository

The MarketMind AI project is maintained in the GitHub repository `Team_2_Small_Biz_AI`. The repository contains the frontend, backend, AI/ML components, datasets, trained models, configuration files, documentation, and supporting project resources.

Table 11.1: GitHub Repository Information

Attribute	Details
Repository Name	<code>Team_2_Small_Biz_AI</code>
Repository Type	Shared team repository
Primary Branch	<code>main</code>
Development Branch	<code>pre-dev</code>
Version Control Platform	GitHub

Repository Structure

The repository follows a modular structure separating the frontend, backend, machine learning components, models, datasets, and supporting resources.

```
Team_2_Small_Biz_AI/  
|  
|-- ML/  
|   |-- customer_segmentation/  
|   |   |-- customer_scaler.pkl  
|   |   |-- customer_segmentation.py  
|   |-- sales_forecasting/  
|       |-- revenue_prediction.py
```

```
|
|-- backend/
|   |-- app/
|   |   |-- core/
|   |   |-- ml/
|   |   |-- routers/
|   |   |-- trained_models/
|   |   |-- main.py
|   |   |-- database.py
|   |   |-- models.py
|   |   |-- schemas.py
|   |-- migrations/
|   |   |-- versions/
|   |-- requirements.txt
|   |-- alembic.ini
|
|-- frontend/
|   |-- public/
|   |   |-- sample-data/
|   |-- scripts/
|   |-- src/
|   |   |-- assets/
|   |   |-- components/
|   |   |-- context/
|   |   |-- hooks/
|   |   |-- pages/
|   |   |-- services/
|   |   |-- utils/
|   |-- package.json
|   |-- vite.config.js
|
|-- models/
|-- sample-data/
|-- System Architecture.drawio
|-- System Architecture.pdf
|-- UI-WIREFRAME-DESIGNS.pdf
|-- preprocessing.py
|-- compress_models.py
|-- requirements.txt
```

| -- README.md

The main repository components are summarised below.

Table 11.2: Major Repository Components

Component	Purpose
frontend/	React-based user interface, pages, dashboards, and reusable components.
backend/	FastAPI services, API routers, database integration, authentication, and AI/ML services.
ML/	Machine learning development code for segmentation, forecasting, and revenue prediction.
models/	Trained machine learning models and supporting artifacts.
sample-data/	Sample datasets used for business and analytical processing.
migrations/	Database schema migration files.
Documentation	Architecture diagrams, wireframes, reports, and project documentation.

Version Control

Git and GitHub were used to manage project changes and support parallel development. The workflow included the **main** branch, **pre-dev** development branch, individual contributor branches, and feature-specific branches such as **recommendation-system**.

The **pre-dev** branch was used for shared development and integration before changes were incorporated into the main project version.



Git commits were used to maintain the history of code changes and support feature integration, debugging, and collaboration. This workflow allowed team members to work independently while keeping the shared MarketMind AI codebase organised.

12. Error Handling & Security

MarketMind AI includes error handling and security mechanisms to improve application reliability and protect user and business data. Since the system integrates frontend, backend, database, and AI/ML components, errors are handled across multiple layers.

12.1 Error Handling

Invalid Input Handling

User inputs and API requests are validated before processing. This helps prevent errors caused by missing fields, incorrect formats, invalid parameters, or unsupported values.

API and Database Errors

The backend handles invalid requests, unavailable resources, processing failures, and database errors. Appropriate responses are returned to the frontend without exposing unnecessary internal system or database details.

AI/ML Error Handling

AI/ML workflows validate required data and model resources before processing. Missing datasets, invalid features, unavailable models, and prediction failures are handled through appropriate error responses.

Frontend Error Feedback

The frontend provides user feedback through validation messages, error notifications, loading indicators, empty states, and retry options where applicable.

Table 12.1: Major Error Handling Considerations

Error Category	Handling Approach
Invalid Input	Validate required fields, formats, and request parameters.
API Errors	Return appropriate responses for failed or invalid requests.
Database Errors	Handle connection and data-operation failures without exposing internal details.
AI/ML Errors	Validate inputs and handle missing models or prediction failures.
Frontend Errors	Display appropriate messages, loading states, and feedback.

12.2 Security

Security mechanisms are used to protect user accounts, business information, application resources, and backend services.

Authentication and Authorization

Users are authenticated before accessing protected functionality. Authorization controls restrict access to resources and operations according to application requirements and user roles.

Password Protection

User passwords are protected using secure password handling and are not exposed through application responses. Where password hashing is implemented, credentials are stored in hashed form rather than plain text.

API Security

Backend APIs validate incoming requests and require authentication for protected endpoints. Sensitive operations are restricted to authorised users.

Sensitive Configuration

Sensitive information such as database connection details, API keys, authentication secrets, and environment-specific settings are managed through environment variables rather than being hard-coded in the source code.

Table 12.2: Security Considerations in MarketMind AI

Security Area	Purpose
Authentication	Verifies user identity before accessing protected functionality.
Authorization	Restricts access to protected resources and operations.
Password Protection	Protects user credentials from direct exposure.
API Security	Validates requests and protects sensitive backend operations.
Configuration Security	Keeps sensitive configuration separate from the source code.

13. Results

This chapter presents the results obtained from the AI/ML components and the overall MarketMind AI system. The results demonstrate the ability of the system to process business data, generate analytical insights, and present the outputs through an integrated user interface.

The results are divided into AI/ML results and system-level results.

13.1 AI/ML Results

The AI/ML components of MarketMind AI were evaluated based on the specific objective of each analytical module. Since the system includes multiple types of machine learning tasks, the evaluation approach varies depending on whether the module performs clustering, forecasting, classification, anomaly detection, or recommendation.

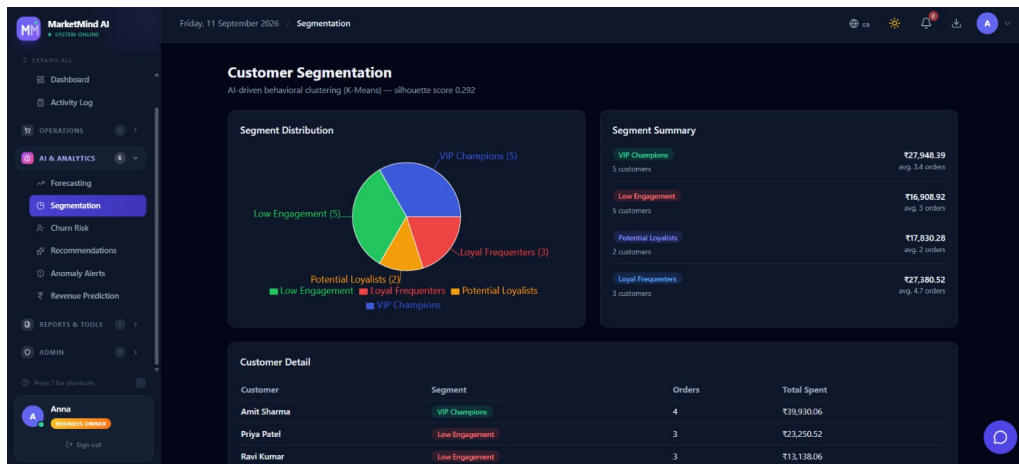
Customer Segmentation

The customer segmentation module grouped customers with similar characteristics and purchasing behaviour into meaningful segments. These segments help businesses understand differences between customer groups and support targeted decision-making.

The generated customer segments can be used to identify groups such as:

- High-value customers.
- Regular customers.
- Customers with lower engagement.
- Customers requiring additional attention.

The segmentation results were evaluated by reviewing the characteristics and business relevance of the generated clusters.



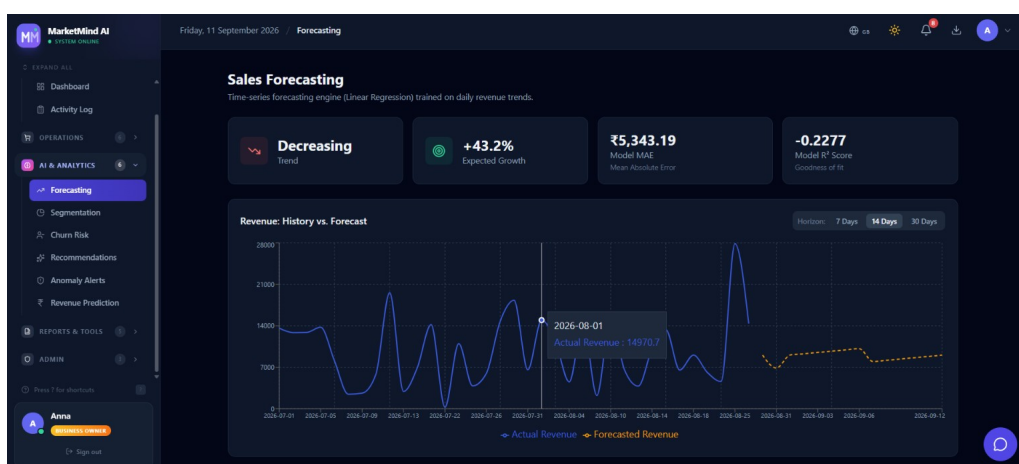
(a) Customer Segmentation Interface

Sales Forecasting and Revenue Prediction

The forecasting module uses historical sales data to estimate future sales and revenue trends, supporting business planning and identifying potential growth or decline.

Table 13.1: Forecasting and Revenue Prediction Results

Metric	Result
Model Used	Random Forest Regressor
MAE	5,343.19
R ² Score	-0.2277



(a) Forecasting Interface

Customer Churn Prediction

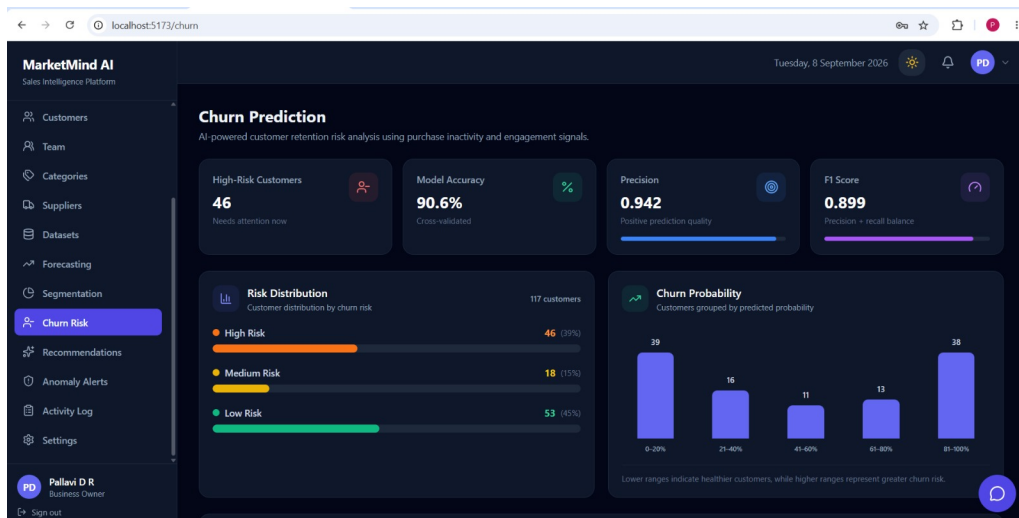
The churn prediction module analyses customer purchase behaviour to identify customers who may be at risk of becoming inactive. The module uses customer activity patterns to estimate churn risk and classify customers into different risk levels.

The results can support customer retention strategies by helping businesses identify customers who may require additional engagement or targeted interventions.

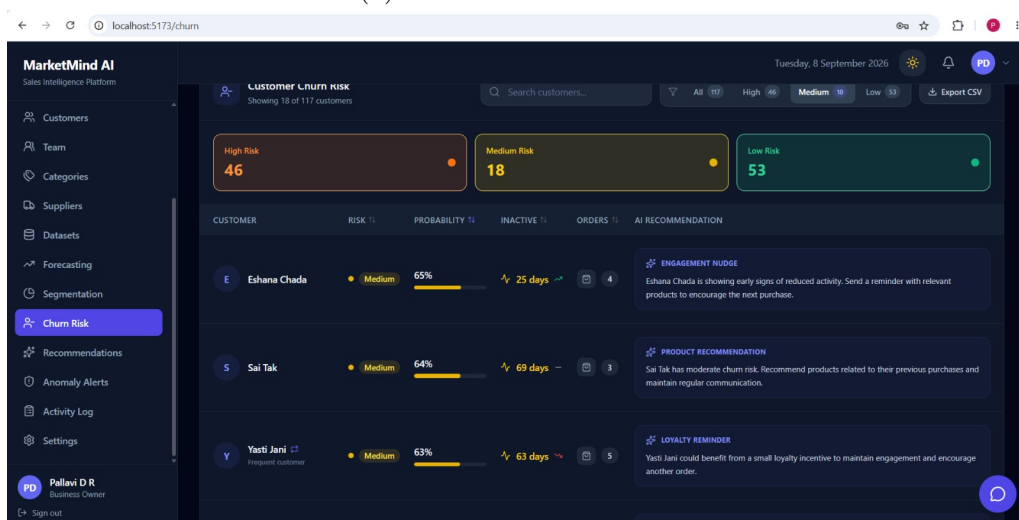
The project specifies Accuracy, Precision, Recall, and F1 Score as the evaluation metrics for the churn prediction model. However, numerical evaluation values were not available in the documented project results. s

Table 13.2: Customer Churn Prediction Evaluation Metrics

Metric	Result
Accuracy	90.6%
Precision	0.942
Recall	0.860
F1 Score	0.899



(a) Churn Risk Overview

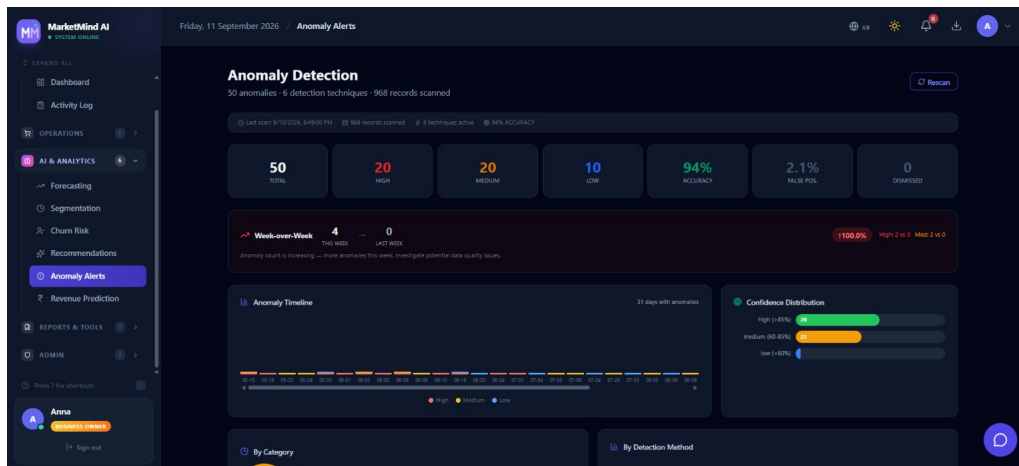


(b) Customer Churn Risk Details

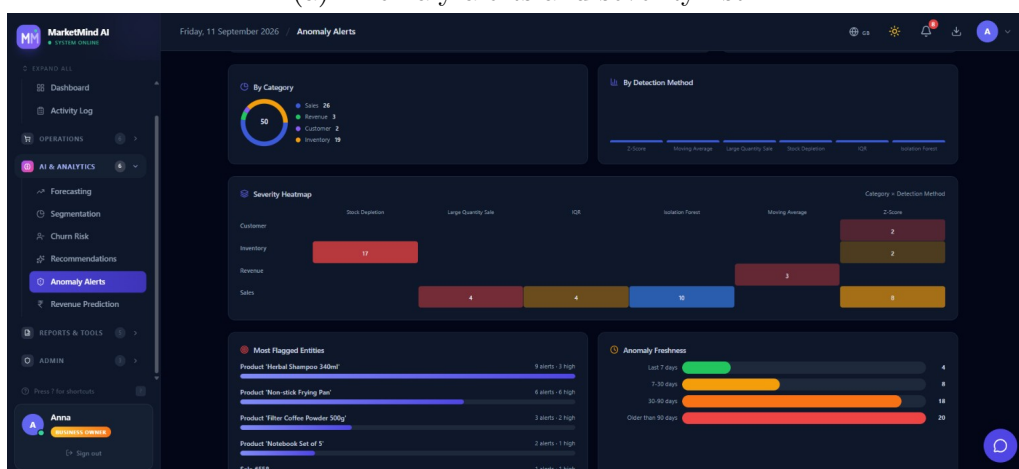
Anomaly Detection

The anomaly detection module identifies unusual patterns or values in business data. These anomalies may represent unexpected changes in sales, transactions, customer behaviour, or other business activities.

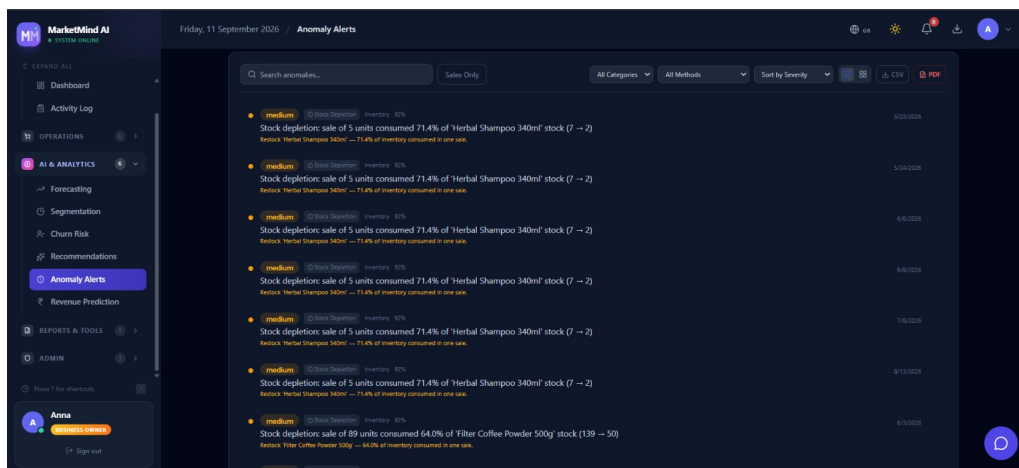
The results help users identify unusual observations that may require further investigation.



(a) Anomaly alerts and severity list



(b) Anomaly analytics and detection analysis



(c) Anomaly detection dashboard

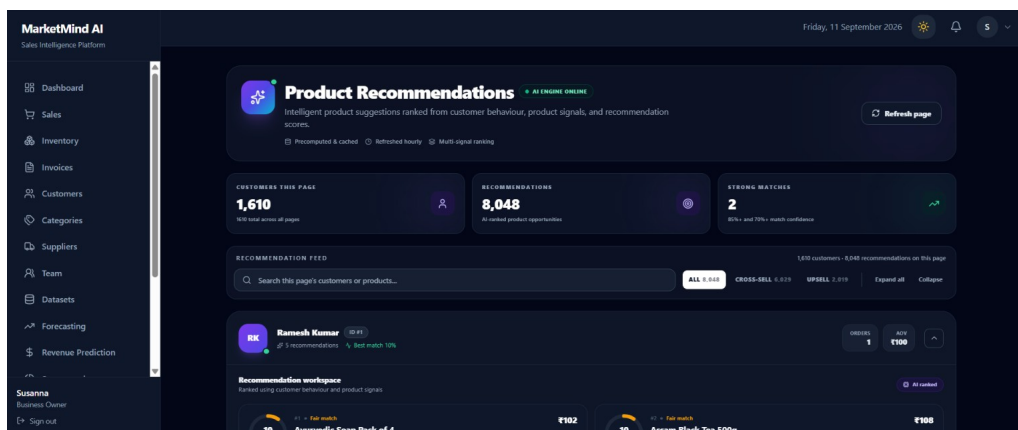
Figure 13.4: Anomaly Detection and Alerts Interface

Product Recommendation Results

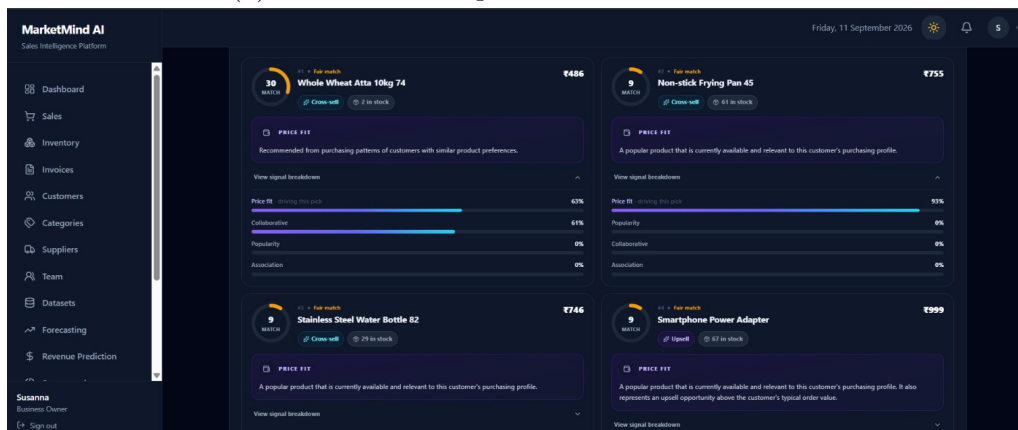
The recommendation module generates product recommendations based on available customer and product information.

The results are intended to help businesses provide relevant product suggestions and improve customer engagement.

Recommendation quality can be evaluated based on the relevance of generated recommendations and the ability of the system to produce appropriate results for the available data.



(a) Customer-level product recommendations



(b) Product recommendation dashboard

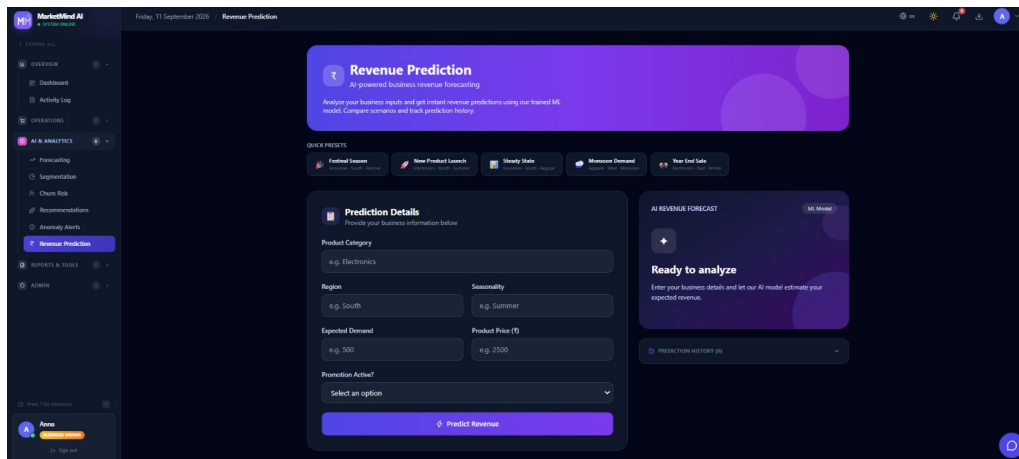
Figure 13.5: MarketMind AI Product Recommendation Interface

Revenue Prediction

The Revenue Prediction module estimates business revenue using the trained machine learning model. Users can provide inputs such as product category, region, seasonality, expected demand, product price, and promotion status.

The interface also provides predefined business scenarios, while the predicted revenue is

displayed through the AI Revenue Forecast panel along with prediction history.



(a) Revenue prediction input interface

Figure 13.6: MarketMind AI Revenue Prediction Interface

AI/ML Results Summary

Table 13.3 summarises the major outputs generated by the AI/ML components.

Table 13.3: AI/ML Results Summary

Module	Primary Output	Evaluation
Customer Segmentation	Customer groups and clusters	Cluster quality and business relevance
Sales Forecasting	Future sales predictions	MAE, RMSE, or other regression metrics
Revenue Prediction	Estimated revenue values	MAE, RMSE, or R^2 score
Churn Prediction	Customer churn risk	Accuracy, Precision, Recall, F1 Score
Anomaly Detection	Unusual observations	Detected anomaly patterns
Recommendations	Suggested products	Recommendation relevance

13.2 System Results

The system-level results demonstrate the integration of the frontend, backend, database, and AI/ML components into a single application.

The implemented system allows users to access business information and analytical insights through a unified dashboard environment.

Application Functionality

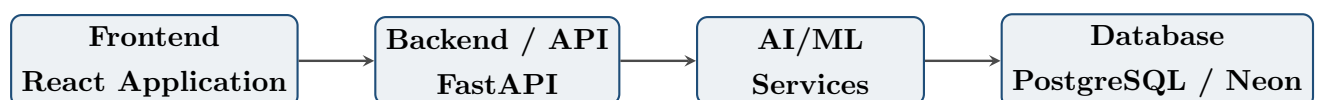
The major implemented functionality includes:

- User authentication and account access.
- Business dashboard and analytics visualisation.
- Customer-related analysis.
- Sales and revenue insights.
- Customer segmentation.
- Sales forecasting.
- Churn prediction.
- Anomaly detection.
- Product recommendations.
- Data-driven business insights.

These features demonstrate the integration of the major components of the MarketMind AI platform.

System Integration Results

MarketMind AI integrates the frontend, backend APIs, AI/ML services, and database into a unified application workflow. Data and requests flow between these components to support business operations and generate analytical insights.



System Performance

System performance was evaluated using the application's built-in performance monitoring and functional testing. The monitoring system records API endpoint execution times, compute time, cache performance, and system activity.

Table 13.4: System Performance Results

Performance Measure	Observed Result
Average API Endpoint Response Time	Approximately 4018 ms for the visible monitored endpoints
Average Compute Time	1325.2 ms
Cache Hit Rate	48%
Performance Monitoring Entries	72 entries recorded
Overall System Functionality	Major application modules operated successfully during functional testing

System Results Summary

The results demonstrate that MarketMind AI successfully brings together business data processing, machine learning analysis, backend services, and user interface functionality within a unified application.

The system provides analytical outputs that can support business users in understanding customer behaviour, identifying trends, generating predictions, and making more informed decisions.

The available results demonstrate the successful implementation of the major project objectives and provide a foundation for further development and improvement.

14. Deployment

MarketMind AI was deployed using a separated frontend and backend architecture. The frontend application is hosted on Vercel, while the FastAPI backend is deployed as a separate service on Render. The application uses PostgreSQL through Neon for persistent data storage.

14.1 Deployment Status

The React frontend was deployed using Vercel, providing hosted access to the web application. The FastAPI backend was deployed on Render as a separate API service.

The deployed frontend communicates with the backend through API requests, while the backend manages application logic, database operations, and AI/ML workflows.

14.2 Deployment Architecture

The deployed system follows a separated frontend, backend, and database architecture.



The frontend provides the user interface and sends requests to the FastAPI backend. The backend handles business logic, database communication, and AI/ML processing before returning results to the frontend.

14.3 Environment Configuration

Deployment-specific configuration is managed using environment variables. Database connection details, API configuration, and other sensitive settings are kept separate from the application source code.

This approach allows the application to use different configurations for development and deployment environments while helping protect sensitive information.

14.4 Deployment Summary

Table 14.1 summarises the deployment configuration of MarketMind AI.

Table 14.1: Deployment Summary

Deployment Area	Technology / Status
Frontend	React application deployed using Vercel.
Backend	FastAPI backend deployed as a separate service on Render.
Database	PostgreSQL database hosted using Neon.
AI/ML Components	Integrated with the FastAPI backend and application workflow.
Application Status	Frontend and backend deployed as separate services.

15. Challenges & Limitations

The development of MarketMind AI involved integrating frontend components, backend APIs, database services, and AI/ML modules. This required continuous debugging, configuration, model integration, and coordination between team members.

Development Challenges

System Integration

Integrating the frontend, backend, database, and AI/ML components into a unified application was a major challenge. Differences in API requests, response formats, data structures, and configuration could affect functionality across multiple layers. Iterative testing and debugging were required to maintain reliable communication between components.

Database Configuration

Database connectivity required correct environment variables and connection settings. Configuration issues could prevent backend services from communicating with the database. This highlighted the importance of separating environment-specific and sensitive configuration from application source code.

AI/ML Model Integration

Integrating machine learning models required careful preprocessing, feature preparation, model input handling, and output formatting. The generated predictions also had to be correctly connected to backend APIs and displayed through the frontend.

Dependency Management

The project uses multiple Python and JavaScript libraries. Maintaining compatible package versions and consistent development environments across team members was necessary to avoid dependency and configuration-related issues.

Team Collaboration and Code Integration

As a team project, different members worked on separate modules and branches. Integrating these contributions required branch management, conflict resolution, and coordination to maintain a consistent application structure.

Current Limitations

Dataset Limitations

The performance of the AI/ML components depends on the quality, size, and diversity of the available datasets. Larger and more diverse datasets could improve model robustness and generalisation.

Model Performance

Model performance can vary depending on feature quality, dataset size, data imbalance, model selection, and hyperparameter configuration. Further experimentation and optimisation could improve predictive performance.

Scalability

The current implementation is primarily designed for project-level usage. Larger datasets and higher numbers of concurrent users may require further database, API, caching, and AI/ML optimisation.

Real-Time Data Processing

The current system primarily works with available business data. Integration with real-time POS, e-commerce, CRM, or other external systems would require additional APIs and data-processing pipelines.

Production Infrastructure

Although the frontend has been deployed using Vercel, scaling the complete system for large-scale production usage would require additional infrastructure, monitoring, security, and performance optimisation.

Summary of Challenges and Limitations

Table 15.1 summarises the major challenges and limitations of MarketMind AI.

Table 15.1: Major Challenges and Current Limitations

Area	Challenge or Limitation
System Integration	Coordinating communication between front-end, backend, database, and AI/ML components.
Database Configuration	Managing database connectivity, environment variables, and configuration.
AI/ML Integration	Preparing model inputs and integrating predictions with application APIs and interfaces.
Dependencies	Maintaining compatible libraries and consistent development environments.
Team Collaboration	Integrating contributions, managing branches, and resolving conflicts.
Dataset	Model performance depends on dataset quality, size, and diversity.
Scalability	Further optimisation may be required for larger datasets and more concurrent users.
Real-Time Processing	External real-time data sources require additional APIs and processing infrastructure.

16. Future Scope

MarketMind AI provides a foundation for combining business analytics, machine learning, and decision support within a unified platform. Future development can focus on improving predictive performance, real-time data integration, personalisation, scalability, security, and production readiness.

Improved AI/ML Models

The AI/ML components can be further improved through better feature engineering, hyperparameter optimisation, additional algorithms, and larger datasets. Improved validation and model comparison could increase the accuracy and reliability of predictive results.

Real-Time Data Integration

Future versions could integrate real-time data from point-of-sale, e-commerce, CRM, inventory systems, and external business APIs. This would allow forecasts, analytics, and business insights to reflect more current information.

Advanced Personalisation and Recommendations

The recommendation system could be enhanced using additional customer behaviour and interaction data. Hybrid recommendation methods, improved ranking algorithms, and real-time recommendation updates could provide more relevant product suggestions.

Expanded Business Analytics

Additional predictive and analytical capabilities could be introduced, including inventory demand prediction, customer lifetime value prediction, dynamic pricing, market trend analysis, sentiment analysis, and advanced revenue optimisation.

Scalability and Performance Improvements

Future development could focus on database optimisation, API performance, caching, AI/ML inference optimisation, and support for larger datasets and concurrent users. Scalable cloud infrastructure could further improve the system's ability to support production-level usage.

Security and System Monitoring

As the platform grows, additional security and monitoring capabilities could be introduced. These may include stronger authentication, enhanced API security, improved protection of sensitive data, performance monitoring, error tracking, and AI/ML model monitoring.

Production Readiness

The frontend is currently deployed using Vercel. Future work can focus on strengthening the complete production infrastructure through scalable backend hosting, automated deployment pipelines, containerisation, monitoring, and improved operational reliability.

Future Scope Summary

The major opportunities for future development are summarised in Table 16.1.

Table 16.1: Future Development Opportunities

Area	Potential Future Improvement
AI/ML	Improve models, features, validation, and prediction performance.
Data Integration	Integrate real-time and external business data sources.
Recommendations	Develop more personalised and dynamic recommendation methods.
Analytics	Add advanced predictive and business intelligence capabilities.
Scalability	Optimise infrastructure for larger datasets and concurrent users.
Security	Strengthen authentication, API security, and data protection.
Monitoring	Add application, API, database, and AI/ML monitoring.
Production	Improve backend hosting, deployment automation, and operational reliability.

17. Conclusion

MarketMind AI was developed as an AI-powered business intelligence and decision-support platform that helps businesses transform available data into meaningful insights.

The system integrates business analytics, machine learning, predictive modelling, and visualisation within a unified application. It provides major analytical capabilities including customer segmentation, sales forecasting, revenue analysis, customer churn prediction, anomaly detection, and product recommendations. These features support businesses in understanding customer behaviour, identifying trends, predicting future outcomes, and recognising potential risks and opportunities.

MarketMind AI was implemented using a full-stack architecture consisting of a React-based frontend, FastAPI backend, PostgreSQL database, and AI/ML components. The integration of these layers enables business data to be processed and transformed into analytical results that can be presented through interactive dashboards and application interfaces.

The development process provided practical experience in full-stack development, machine learning integration, database configuration, debugging, dependency management, version control, and collaborative software development. The major application modules were successfully integrated and tested, and the frontend was deployed using Vercel.

Although the current implementation has limitations related to dataset availability, model performance, scalability, and real-time data processing, the platform provides a strong foundation for further development. Future improvements can include real-time data integration, improved AI/ML models, advanced personalisation, expanded analytics, stronger security, and scalable production infrastructure.

Overall, MarketMind AI demonstrates the practical application of artificial intelligence and modern web technologies to business intelligence and data-driven decision-making. The project establishes a foundation that can be further enhanced to provide more accurate, personalised, and scalable business insights.

Final Project Summary

MarketMind AI brings together the following key capabilities:

- Business data analysis and visualisation.

- Customer segmentation and behavioural insights.
- Sales forecasting and predictive analytics.
- Revenue and business performance analysis.
- Customer churn prediction.
- Anomaly detection.
- Product recommendations.
- Full-stack application development.
- AI/ML model integration.
- Data-driven decision support.

The project demonstrates a practical implementation of an intelligent business analytics platform and provides a foundation for continued development and production-level enhancement.

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